ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN
(QSP)

WEAPONS ENGINEERING TECHNICIAN
(W ENG TECH - 00366)

USQ – S1 W Eng Tech Maintainer Communications (AKQH)

MITE COURSE ID:
Residential Phase - 118941
DRMIS Phase – 119097
MSCU Phase – 121299
NMT Phase – 121114
TRS Phase – 120591

Regular Force: Weapons Engineering Technician 0036 (W ENG Tech)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): NFS(P)

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Approval Date	Authority
1.0	21 Feb 11	DMTE 2-4-2
xx	xxxxxxx	xxxxx

NOTE:
Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the NTG Website. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the following:

a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-002 Non-Commission Member General Specification dated 09 February 2011; and

c. A-PD-055-002/PP-001 Weapons Engineering Technician 00366 dated 01 September 2011.

3. This QSP publication is effective on receipt and supersedes AKQR QSP revision 1.2 dated Jul 2014 (RDIMS #XXXXX). The QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. This QS at Chapter 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 15 Jan 2023 working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened Apr 2023 at NFS(P) working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outline in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NPTORD 4500-22 is required.

QUALIFICATION STANDARD

CHAPTER 1: QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the W Eng Tech to perform their duties as a Communications Maintainer. Possession of this qualification will allow personnel to operate, troubleshoot, and perform preventative and corrective maintenance on communications equipment.

GRANTING OF NQUAL

2. A trainee will achieve the qualification AKQH (MITE 118941) and W Eng Tech Communications Maintainer when the standards listed in the Performance Objectives (POs) have been met.

a. Communications Applications, MITE 118941

b. MSCU, MITE 121299.

c. NMT, MITE 121114

d. TRS, MITE 120591

CO-QUALIFICATION

3. In Addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. DRMIS, MITE 119097.

b. Specialist Technical Inspector MITE 302486

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. AKNV_ Jr W Eng Tech EQ2 MITE 118939 or MITE 123762; and

b. AKQH_W Eng RQS1 Academics MITE 118941

CHAPTER 2: PERFORMANCE OBJECTIVES

PO 001 Maintain Internal Communications System

1. Performance. Maintain internal communications system (SHINCOM).

2. Conditions

    (1) Given:

    (2) supervision;

    (3) references;

    (4) DWAN;

    (5) tools and test equipment;

    (6) PPE;

    (7) Internal Communications System; and

    (8) assistance.

    b. Denied:

    (1) Nil.

    c. Environmental:

    (1) all shipboard conditions at sea or alongside; and

    (2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain the internal communications system, IAW refs, by:

    a. manipulating the SHINCOM;

    b. maintaining the SHINCOM;

    c. describing the Trunked Radio System;

    d. initializing the Trunk Radio System; and

    e. maintaining the Trunk Radio System.

4. References:

    a. A1 – A7; and

b. A9 – A21.

5. Task Numbers.

a. I001 – I015.

6. Training Limitations:

a. Nil.

PO 002 Maintain UHF Communication System

1. Performance. Maintain UHF communication system.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) UHF Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain UHF communication system, IAW refs, by:

   a. operating the UHF communication systems, to include:

      (1) powering up;

      (2) carrying out system functional checks;

      (3) verifying remote operation;

   b. performing CM on the UHF communication systems, to include:

      (1) identifying deficiency;

      (2) performing diagnostics;

      (3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU; and

(6) performing functional check;

c. completing maintenance administration.

4. References:

a. A1 – A7;

b. A8; and

c. A22 – A38.

5. Task Numbers:

a. J001 – J012.

6. Training Limitations:

a. Nil.

PO 003 Maintain HF Transmit System

1. Performance. Maintain the HF transmit system.

2. Conditions:

a. Given:

(1) supervision;

(2) references;

(3) DWAN;

(4) tools and test equipment;

(5) PPE;

(6) HF Transmit System; and

(7) assistance.

b. Denied:

(1) Nil.

c. Environmental:

(1) all shipboard conditions at sea or alongside; and

(2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain the HF transmit system, IAW refs, by:

a. operating the HF transmit system, to include:

(1) powering up.

b. carrying out system functional checks, to include:

(1) verifying remote operation.

c. performing PM on the HF Transmit System, to include:

(1) interpreting CFTO;

(2) performing PM IAW CFTO; and

(3) completing maintenance administration.

d. performing CM on the HF transmit system, to include:

(1) identifying deficiency;

(2) performing diagnostics;

(3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU;

(6) performing functional check; and

e. completing maintenance administration.

4. References:

a. A1 – A7;

b. A8; and

c. A39 – A56.

5. Task Numbers:

a. J013 – J024.

6. Training Limitations:

a. Nil.

PO 004 Maintain LF/MF/HF Receive System

1. Performance. Maintain the LF/MF/HF receive system

2. Conditions:

a. Given:

(1) supervision;

(2) references;

(3) DWAN;

(4) tools and test equipment;

(5) PPE;

(6) LF/MF/HF Receive System; and

(7) assistance.

b. Denied:

(1) Nil.

c. Environmental:

(1) all shipboard conditions at sea or alongside; and

(2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain the LF/MF/HF receive system, IAW refs, by:

a. operating the LF/MF/HF receive system, to include:

(1) powering up;

(2) carrying out system functional checks; and

(3) verifying remote operation.

b. performing PM on the LF/MF/HF receive system, to include

(1) interpreting CFTO;

(2) performing PM IAW CFTO; and

(3) completing maintenance administration.

c. performing CM on the LF/MF/HF receive system, to include

(1) identifying deficiency;

(2) performing diagnostics;

(3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU;

(6) performing functional check; and

(7) completing maintenance administration.

4. References:

a. A1 – A7;

b. A8; and

c. A57 – A73.

5. Task Numbers:

a. J025 – J036.

6. Training Limitations:

a. Nil.

PO 005 Maintain Communications Control and Monitoring System (CCMS)

1. Performance. Maintain the Communications Control and Monitoring System (CCMS)

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CCMS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain the CCMS, IAW refs, by:

   a. operating the CCMS, to include:

   b. powering up;

   c. carrying out system functional checks, to include:

   d. verifying remote operation.

   e. performing PM on the CCMS, to include:

      (1) interpreting CFTO;

      (2) performing PM IAW CFTO; and

(3) completing maintenance administration.

f. Initialize a communication circuit utilizing the black patch panel, to include:

(1) plain voice;

(2) secure voice; and

(3) link.

4. References:

a. A1 – A7;

b. A8; and

c. A74 – A88.

5. Task Numbers:

a. L001 – L011.

6. Training Limitations:

a. Nil.

PO 006 Maintain the UHF Satellite Communication System

1. Performance. Maintain the UHF Satellite communication system.

2. Conditions:

(1) Given:

(2) supervision;

(3) references;

(4) DWAN;

(5) tools and test equipment;

(6) PPE;

(7) UHF Satellite Communications System; and

(8) assistance.

b. Denied:

(1) Nil.

c. Environmental:

(1) all shipboard conditions at sea or alongside; and

(2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain the UHF satellite communications system, IAW refs, by:

a. operating the UHF Satellite Communications System, to include:

(1) powering up;

(2) carrying out system functional checks; and

(3) verifying remote operation.

b. performing PM on the UHF Satellite Communications System, to include:

(1) interpreting CFTO;

(2) performing PM IAW CFTO; and

(3) completing maintenance administration.

c. performing CM on the UHF Satellite Communications System, to include:

(1) identifying deficiency;

(2) performing diagnostics;

(3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU;

(6) performing functional check; and

(7) completing maintenance administration.

4. References:

a. A1 – A7;

b. A8;

c. A22 – A38;

d. A89 – A96; and

e. A97 – A102.

5. Task Numbers:

a. J037 – J048.

6. Training Limitations:

a. Nil.

PO 007 Maintain the Satellite Communication System

1. Performance. Maintain the satellite communication system.

2. Conditions:

a. Given:

(1) supervision;

(2) references;

(3) DWAN;

(4) tools and test equipment;

(5) PPE;

(6) Satellite Communications System; and

(7) assistance.

b. Denied:

(1) Nil.

c. Environmental:

(1) all shipboard conditions at sea or alongside; and

(2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain the satellite communications system, IAW refs, by:

a. operating the Satellite Communications System, to include:

(1) powering up;

(2) carrying out system functional checks; and

(3) verifying remote operation.

b. performing PM on the Satellite Communications System, to include:

(1) interpreting CFTO;

(2) performing PM IAW CFTO; and

(3) completing maintenance administration.

c. performing CM on the Satellite Communications System, to include:

(1) identifying deficiency;

(2) performing diagnostics;

(3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU; and

(6) performing functional check.

d. completing maintenance administration.

4. References:

a. A1 – A7;

b. A8;

c. A98 – A102; and

d. C6.

5. Task Numbers:

a. K001 – K010.

6. Training Limitations:

a. Nil.

PO 008 Not Allocated

PO 009 Maintain Tactical Data System

1. Performance. Maintain the tactical data system.

2. Conditions:

a. Given:

(1) supervision;

(2) references;

(3) DWAN;

(4) tools and test equipment;

(5) PPE;

(6) Multi-Link and peripherals;

(7) software and crypto if needed; and

(8) assistance.

b. Denied:

(1) Nil.

c. Environment

(1) any shipboard conditions at sea or alongside; and

(2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain the Tactical Data System, IAW refs, by;

a. operating the Tactical Data System, to include:

(1) powering up;

(2) carrying out system functional checks; and

(3) verifying remote operation.

b. performing PM on the Tactical Data System, to include:

(1) interpreting CFTO;

(2) performing PM IAW CFTO; and

c. completing maintenance administration. Performing CM on the Tactical Data System, to include:

(1) identifying deficiency;

(2) performing diagnostics;

(3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU;

(6) performing functional check.

d. completing maintenance administration.

4. References:

a. A1 – A7;

b. A8;

c. A112 – A115;

d. A116 – A126;

e. C5; and

f. C25 – C30.

5. Task Number:

a. 0001 – 0011.

6. Training Limitations:

a. Nil.

PO 010 Maintain Combat Management System

1. Performance. Maintain Combat Management System (CMS).

2. Conditions:

a. Given:

(1) supervision;

(2) references;

(3) DWAN;

(4) tools and test equipment;

(5) PPE;

(6) CMS / MPT; and

(7) assistance.

b. Denied:

(1) Nil.

c. Environmental:

(1) any shipboard conditions at sea or alongside; and

(2) day or night.

3. Standard. The W Eng Tech Communications System Maintainer shall maintain the CMS, IAW refs, by:

a. operating the CMS, to include:

(1) power up;

(2) carry out system functional checks; and

(3) verify remote operation.

b. perform PM on the CMS, to include:

(1) interpret CFTO;

(2) perform PM IAW CFTO; and

(3) complete maintenance administration.

c. perform CM on the CMS, to include:

(1) identify deficiency;

(2) perform diagnostics;

(3) conduct fault finding;

(4) identify faulted LRU;

(5) replace LRU; and

(6) perform functional checks.

d. complete maintenance administration.

4. References:

a. A1 – A7; and

b. A112 – A115.

5. Task Numbers:

a. 0012 – 0022.

6. Training Limitations:

a. Nil.

PO 011 Utilize DRMIS Portal as a Maintenance Supervisor (MOD 2)

1. Performance: Utilize DRMIS Portal as a Maintenance Supervisor (Mod 2) IAW PO 002 of DRMIS QSP 119097.

2. Conditions:

   a. Given:

      (1) references;

      (2) supervision;

      (3) computer with LAN access;

      (4) printer;

      (5) DRMIS account; and

      (6) access to DRMIS Portal.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any condition at sea or alongside; and

      (2) day or night.

3. Standard: Utilize DRMIS Portal as a Maintenance Supervisor (Mod 2) IAW PO 002 of DRMIS QSP 119097.

4. References:

   a. IAW PO 002 of the DRMIS QSP.

5. Specification Task Numbers:

   a. N/A.

6. Training Limitations:

   a. Nil.

TRAINING PLAN

CHAPTER 3: TRAINING MANGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. Enabling Checks (ECs) are formative tests that help both Trainee and instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and Trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the Trainee's final mark;

   b. Performance Checks (PCs) are summative tests used to determine if the Trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Summative tests can take the form of PCs or Critical ECs (reserved for safety related content). Supplemental PCs are administered upon failure of a PC or critical EC and must be in the same format and have the same standard as the original PC; and

   c. Use of Applicable Training Simulator software.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 122 training days.

Actual Training Time (ATT)	XXX periods
Training Administrative Time (TAT)	XXX periods
Base Administrative Time (BAT)	XXX periods
Training Travel Time (TTT)	XXX
Course Duration (Total Calendar Days) (ATT+TAT+BAT+TTT)/8	XXX training days

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Course Director

Rank	MOSID	Qualifications Required	Qty	PO / EO

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
MS/PO2	00366	W ENG Tech Communications Applications, Bit	2	All
PO1	00366	W ENG Tech Maintenance Manager	1	All

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

   a. Limitation 1; and
   b. Limitation 2.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

   a. Optimum – 1:6;
   b. Maximum Load – 1:8; and
   c. Minimum –1:2.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

   a. The Career Manager for career courses; or
   b. NFS Campus Support for occupation and specialty courses.

QUALIFICATION and ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-22. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 and/or CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4: LESSON SPECIFICATIONS

PO 001 Maintain Internal Communications System

EO 001.01 Manipulate the SHINCOM

1. Performance. Manipulate the SHINCOM.

2. Conditions;

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Internal Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall manipulate the SHINCOM Switch, IAW refs, by:

   a. describing the following, to include:

      (1) system operation;

      (2) system configurations;

      (3) system functions;

      (4) physical layout of equipment; and

      (5) indicators and controls.

b. manipulating the SHINCOM software

4. Time Required:

a. EO - 14 x 50 min periods.

5. Instructional Methods:

a. lecture;

b. demonstration; and

c. practical.

6. Training Aids:

a. SHINCOM and/or MPT;

b. audio visual aids; and

c. handouts.

7. Test Details:

a. EO 001.01 will be tested as a part of PC 001 IAW Annex K.

8. References.

a. A1 – A7; and

b. A9 – A21.

9. Remarks:

a. projects and exercises will be assigned; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. System Operation by describing; a. system operation; b. modes of operation; and c. capabilities and limitations	L		
TP2	1. System configuration by describing; a. system block diagram; and b. general signal flow.	L		
TP3	1. System functions by describing the function of the console group to include; a. Describe the CSU; b. Control Node; c. Concentrator Node; d. Console group configuration; e. Describe the MDF/Junction Box; f. The Terminal Equipment;	L		

	g. The interfaces internal CCA; h. Function of the power monitoring board distribution; and i. Maintenance console.			
TP4	1. Physical layout of equipment by describing to include: a. ships physical arrangement b. cabinet layout	D		
TP5	1. Indicators and controls to include: a. familiarize with publication layout; b. familiarize with location of controls and indicators; c. describe built in test equipment to include: d. sanity check; and e. continuity check	L		
TP6	1. Manipulate the SHINCOM	P		

PO 001 Maintain Internal Communications System

EO 001.02 Maintain the SHINCOM

1. Performance. Maintain the SHINCOM.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Internal Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the SHINCOM, IAW refs, by:

   a. localizing fault to LRU;

   b. replacing and aligning LRU; and

   c. verifying operational readiness.

4. Time Required:

   a. EO - 16 x 50 min periods; and

   b. PC – 8 x 50 min periods.

5. Instructional Methods:

a. practice; and

b. PC

6. References:

a. A1 – A7; and

b. A9 – A21.

7. Training Aids:

a. SHINCOM and/or MPT;

b. audio visual aids;

c. handouts; and

d. references.

8. Test Details:

a. EO 001.02 will be tested as part of PC 001 with a time limit of 45 min IAW Annex K

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned;

c. assignments are to be completed during non-instructional time; and

d. PC 001 is 1 X 50 min period with corrective measures starting at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Localize fault to LRU by conducting faultfinding to include: a. investigate the SHINCOM; b. observe device status on the Maintenance Console; c. utilizing logical procedure; and d. restoring faulty device to operational state.	P	16	
TP2	1. Replace LRU to include: a. alignment			
TP3	1. Verifying operational readiness to include: a. verify BITE results on the Maintenance Console; and b. operationally test the system			

PO 001 Maintain Internal Communications System

EO 001.03 Describe the Trunked Radio System

1. Performance. Describe the Trunked Radio System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Trunked Radio System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the Trunked Radio System, IAW refs, by:

   a. System operation;

   b. System configurations;

   c. System functions;

   d. Physical layout of equipment; and

   e. Indicators and controls.

4. Time Required:

   a. EO - 7 x 50 min periods.

5. Instructional Methods:

   a. Lecture.

6. References:

   a. LEONARDO DRS Internal Communications Trunked Radio System Technical Manual, MAN-1199, Revision D

7. Training Aids:

   a. Audio visual aids; and

   b. Handouts.

8. Test Details:

   a. EO 001.03 will be tested in PC 001.

9. Remarks:

   a. Projects and exercises may be assigned; and

   b. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe the Trunked Radio System to include: a. System operation; b. Modes of operation; and c. Capabilities and limitations.	L	1	
TP2	1. Describe the system configuration to include: a. System block diagram; and b. General signal flow.	L	1	
TP3	1. Describe the system functions to include: a. Describe the ICTRS Cabinet; b. Base Station; c. Tetra IP Node; d. Network management Server; e. DAS; f. The Handheld Portable Radio; g. The Fixed Base Radio; h. SHINCOM Interface; i. Power Distribution; and	L	4	

	j. Maintainer Console.			
TP4	1. Describe the physical layout of equipment to include: a. Ships physical arrangement; b. Cabinet layout; and c. Antenna location.	L	0.5	
TP5	1. Describe the indicators and controls to include: a. Publication layout; b. Locations of controls and indicators; and c. Built in test equipment.	L	0.5	

PO 001 Maintain Internal Communications System

EO 001.04 Initialize the Trunked Radio System

1. Performance: Initialize the Trunked Radio System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Trunked Radio System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize the Trunked Radio System, IAW refs, by:

   a. Powering up;

   b. Carrying out equipment functional checks;

   c. Verifying operational checks;

   d. Reporting equipment status to supervisor; and

   e. Powering down.

4. Time Required:

   a. EO - 14 x 50 min periods.

5. Instructional Methods:

   a. Demonstration; and

   b. Practice.

6. References:

   a. LEONARDO DRS Internal Communications Trunked Radio System Technical Manual, MAN-1199, Revision D

7. Training Aids:

   a. ICTRS with system interfaces;

   b. Handouts; and

   c. Audio visual aids.

8. Test Details:

   a. EO 001.04 will be tested in PC 001.

9. Remarks:

   a. Projects and exercises may be assigned; and

   b. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize the Trunked Radio System by powering up, to include: a. NMS; b. AC Controller Unit; c. DC Distribution Unit; d. Enable Forward Base Station RFU; e. Fixed Radio; and f. UPS.	P	8	
TP2	1. Carry out equipment functional checks, to include: a. Powering & login into the Maintainer Console; b. Verify alarms; and clear all alarms.			
TP3	1. Verify operational readiness by performing a radio check.			
TP4	1. Report equipment status to supervisor.	P	1	
TP5	1. Perform power down	P	4	

PO 001 Maintain Internal Communications System

EO 001.05 Maintain the Trunked Radio System

1. Performance: Maintain the Trunked Radio System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Trunked Radio System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the Trunked Radio System, IAW refs, by:

   a. Localizing fault to LRU;

   b. Replacing and aligning LRU; and

   c. Verifying operational readiness.

4. Time Required:

   a. EO - 49 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

b. Demonstration; and

c. Practice.

6. References:

a. LEONARDO DRS Internal Communications Trunked Radio System Technical Manual, MAN-1199, Revision D

7. Training Aids:

a. Handouts; and

b. Audio visual aids.

8. Test Details:

a. EO 001.05 will be tested in PC 001.

9. Remarks:

a. Projects and exercises may be assigned; and

b. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Localize fault to LRU by conducting faultfinding to include: a. Initialize the TRS; b. Observe fault indicators; c. utilizing logical procedure; and d. restoring faulty device to operational state.	P	1	
TP2	1. Repairing and aligning system to include: a. Replace faulted LRU; and b. Perform alignment as necessary.	P	28	
TP3	1. Verify operational readiness to include: a. Verify alarms on the Maintainer Console; and b. Operationally test the system	P	5	

PO 002 Maintain UHF Communication System

EO 002.01 Describe the UHF Communication System

1. Performance. Describe the UHF Communication Systems.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) UHF Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the UHF Communication Systems, IAW refs, by:

   a. system operation;

   b. system configurations;

   c. system functions;

   d. physical layout of equipment; and

   e. indicators and controls.

4. Time Required:

   a. EO – 5 x 50 min periods.

5. Instructional Methods:

   a. Lecture; and

   b. demonstration.

6. References:

   a. A1 – A7;

   b. A8; and

   c. A22 – A38.

7. Training Aids:

   a. an UHF Communication System and/or MPT;

   b. audio visual aids; and

   c. handouts.

8. Test Details:

   a. EO 002.01 will be tested as part of PC 002 IAW Annex K.

9. Remarks:

   a. projects and exercises may be assigned; and

   b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe the UHF Communication System to include: a. system operation; b. modes of operation; and c. capabilities and limitations	L	0.5	
TP2	1. Describe the system configuration to include: a. system block diagram; and b. general signal flow	L	0.5	
TP3	1. Describe the system functions to include: a. transceiver; b. coupler; c. antenna; d. air dryer; and e. guard receiver	L	2	
TP4	1. Describe the physical layout of equipment to include: a. ships physical arrangement;	D	1	

	b. cabinet layout; and c. antenna location.			
TP5	1. Describe the physical layout of equipment to include: a. ships physical arrangement; b. cabinet layout; and c. antenna location.	D	1	
TP6	1. Describe the indicators and controls to include: a. publication layout b. locations of controls and indicators; and c. built in test equipment	L D	0.5 0.5	

PO 002 Maintain UHF Communication System

EO 002.02 Initialize UHF Communication System

1. Performance. Initialize the UHF Communication System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) UHF Communication System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize the UHF communication System, IAW refs, by:

   a. powering up;

   b. carrying out equipment functional checks;

   c. verifying operational checks;

   d. reporting equipment status to supervisor; and

   e. powering down.

4. Time Required:

   a. EO - 4 x 50 min periods.

5. Instructional Methods:

   a. demonstration; and

   b. practice.

6. References:

   a. A1 – A7;

   b. A8; and

   c. A22 – A38.

7. Training Aids:

   a. UHF Communication Systems with system interfaces and/or MPT;

   b. handouts; and

   c. audio visual aids.

8. Test Details:

   a. EO 002.02 will be tested as part of PC 002 IAW Annex K.

9. Remarks:

   a. MPT may be utilized,

   b. a signal flow project may be assigned;

   c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize the UHF Communication System by powering up, to include: a. verify that all controls and switches are in the correct position; and b. perform power on.	D P	0.5 3.5	
TP2	1. Carry out equipment functional checks, to include: a. perform BITE tests b. verify air pressure; and c. verify coupler ready light			
TP3	1. Verify operational readiness by performing a radio check.			
TP4	1. Report equipment status to supervisor.			
TP5	1. Perform power down.			

PO 002 Maintain UHF communication System

EO 002.03 Maintain UHF Transceivers

1. Performance. Maintain the UHF Transceivers.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) UHF Communication System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the UHF Transceivers, IAW refs, by:

   a. describing the UHF Transceiver function;

   b. localizing fault to LRU;

   c. repairing and aligning system; and

   d. verifying operational readiness;

4. Time Required:

   a. EO – 21 x 50 min periods.

5. Instructional Methods:

a. lecture; and

b. practice.

6. References:

a. A1 – A7;

b. A8; and

c. A22 – A38.

7. Training Aids:

a. UHF System and/or MPT;

b. audio visual aids; and

c. handouts;

8. Test Details:

a. EO 002.03 will be tested as part of PC 002 IAW Annex K.

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe the UHF Transceiver function, to include: a. describe overall WSC-3 function; b. reproduce WSC-3 block diagram; c. identify and describe sub-assemblies functions, to include; 1) WSC-3 modules A1A1 through A1A25; 2) RT-1107 specific modules; and d. describe the capabilities and limitations of the equipment.	L	13	
TP2	1. Localizing fault to LRU to include: a. Initialize the UHF System; b. observe fault indicators; c. perform and interpret BITE test; d. utilize trouble-shooting flow charts;	P	8	

	e. measure test points as necessary identify faulted LRU.			
TP3	1. Repairing and aligning system to include: a. replace faulted LRU; and b. perform alignment as necessary. c. Verify operational readiness to Include: 1) perform and interpret BITE test 2) perform radio check			

PO 002 Maintain UHF Communication System

EO 002.04 Maintain the UHF Coupler Group

1. Performance. Maintain the UHF Coupler Group.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) UHF Communication System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the UHF coupler Group, IAW refs, by:

   a. describing the UHF Coupler Group;

   b. localizing fault to LRU;

   c. repairing and aligning system; and

   d. verifying operational readiness.

4. Time Required:

   a. EO – 26 x 50 min periods; and

   b. PC – 8 x 50 min periods.

5. Instructional Methods:

a. lecture;

b. practice; and

c. PC.

6. References:

a. A1 – A7;

b. A8; and

c. A22 – A38.

7. Training Aids:

a. UHF System and/or MPT;

b. audio visual aids;

c. handouts; and

d. references.

8. Test Details:

a. EO 002.04 will be tested as part of PC 002;

b. PC 002 and debrief to be conducted IAW Annex K with a 60 min time limit; and

c. the PC will consist of a minimum of 1 power fault, 1 coupler fault, and 1 WSC- 3 fault.

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned;

c. assignments are to be completed during non-instructional time; and

d. eight practice periods are allotted for overall system faultfinding prior to PC 002.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe the UHF Coupler Group. to include: a. describe overall OE-5012 function; b. reproduce OE-5012 block diagram; and c. describe controls and indicators. 2. Identify and describe sub-assemblies functions, to include: a. CU-5119 Coupler A1 to A6 CCA; b. Guard Coupler; c. Combiner; and d. Power meter. 3. Describe capabilities and limitations of the equipment.			
TP2	1. Localizing fault to LRU by: a. initialize the UHF System; b. observe fault indicators; c. utilize trouble-shooting flow charts;			

	d. measure test points as necessary; and e. identify faulted LRU.			
TP3	1. Repairing and aligning system by: a. replace faulted LRU; and b. perform alignment as necessary .			
TP4	1. Verify operational readiness by tuning the coupler.			

PO 003 Maintain HF Transmit System

EO 003.01 Describe the HF Transmit System

1. Performance. Describe the HF Transmit System.

2. Conditions.

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) HF Transmit System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the HF transmit System, IAW refs, to include:

   a. system operation;

   b. system configurations;

   c. system functions;

   d. physical layout of equipment; and

   e. indicators and controls.

4. Time Required.

   a. EO - 10 x 50 min periods.

5. Instructional Methods.

   a. lecture; and

   b. demonstration.

6. References.

   a. A1 – A7;

   b. A8; and

   c. A39 – A56.

7. Training Aids.

   a. HF Communication System and/or MPT;

   b. audio visual aids; and

   c. handouts.

8. Test Details.

   a. EO 003.01 will be tested as part of PC 003 IAW Annex K.

9. Remarks.

   a. projects and exercises may be assigned; and

   b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. HF Transmit System operation by describing: a. system operation; b. modes of operation; and c. capabilities and limitations.	L		
TP2	1. System configurations by describing: a. system block diagram; b. general signal flow.	L	1	
TP3	1. System function by describing, to include: a. TSCU; b. ISDU; c. Band pass filter; d. OCU; e. TSM; f. 4kW attenuator; g. EMCON relay assembly; h. RTCU; i. Antenna group; and j. Coupler group;	L	4	

	2. Fault-locating indicator and be familiar with: a. exciter b. RF power supply; and c. RF amplifier;			
TP4	1. Physical layout of equipment by describing, to include: a. ships physical arrangement; b. cabinet layout; and c. antenna location.	D	2	
TP5	1. Indicators and controls to include: a. familiarize with publication layout; b. familiarize with location of controls and indicators; c. describe Built in test equipment; and d. describe power meter.	D	2	

PO 003 Maintain the HF Transmit System

EO 003.02 Initialize the HF Transmit System

1. Performance. Initialize the HF transmit system.

2. Conditions.

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) HF Transmit System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize the HF transmit system, IAW refs, by:

   a. powering up;

   b. carrying out equipment functional checks;

   c. verifying operational readiness;

   d. reporting equipment status to supervisor; and

   e. powering down.

4. Time Required.

   a. EO - 3.5 x 50 min periods.

5. Instructional Methods.

   a. demonstration; and practice.

6. References.

   a. A1 – A7;

   b. A8; and

   c. A39 – A56.

7. Training Aids.

   a. HF transmit system and/or MPT;

   b. handouts; and

   c. audio visual aids.

8. Test Details.

   a. EO 003.02 will be tested as a part of PC 003 IAW Annex K.

9. Remarks.

   a. MPT may be utilized;

   b. projects and exercises may be assigned; and

   c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize the HF Transmit System by: a. verify that all controls and switches are in the correct position; and b. perform Power up in the correct sequence.	D P	1 2.5	
TP2	1. Carry out equipment functional checks by: a. perform and interpret BITE tests; and b. verify forward and reflected power.			
TP3	1. Verify operational readiness by performing radio check.			
TP4	1. Report equipment status to supervisor.			
TP5	1. Perform Power down in the correct sequence.			

PO 003 Maintain the HF Transmit System

EO 003.03 Maintain the ISDU and TSCU

1. Performance. Maintain the ISDU and TSCU.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) HF Transmit System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the ISDU and TSCU, IAW refs, by:

   a. describing the ISDU and TSCU

   b. localizing fault to LRU;

   c. repairing and aligning system; and

   d. verifying operational readiness.

4. Time Required:

   a. EO - 22 x 50 min periods.

5. Instructional Methods:

a. lecture; and

b. practice.

6. References:

a. A1 – A7;

b. A8; and

c. A39 – A56.

7. Training Aids.:

a. HF transmit system with system interfaces and/or MPT;

b. audio visual aids; and

c. handouts.

8. Test Details:

a. EO 003.03 will be tested as a part of PC 003 IAW Annex K.

9. Remarks:

a. MPT may be utilized;

b. a signal flow project may be assigned; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Maintain the ISDU and TSCU, to include: a. describe overall ISDU function; b. describe overall TSCU function; c. reproduce block diagram for ISDU; d. describe controls and indicators for ISDU; e. describe controls and indicators for TSCU f. identify and describe sub-assemblies functions for ISDU; and g. describe capabilities and limitations of the equipment	L	14	
TP2	1. Localizing fault to LRU, to include: a. initialize the HF Transmit System; b. observe fault indicators; c. utilize trouble-shooting flow charts; and d. identify faulted LRU.			

TP3	1. Repairing and aligning system, to include: a. replace faulted LRU; and b. verify DIP switch position.			
TP4	1. Verify operational readiness, to include: a. perform full system test; and b. power down HF Transmit System.			

PO 003 Maintain the HF Transmit System

EO 003.04 Maintain the HF Transmit System

1. Performance. Maintain the HF Transmit System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) HF Transmit System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the HF transmit System, IAW refs, by describing sub-assembly function to include:

      (1) band pass filter;

      (2) OCU;

      (3) TSM;

      (4) 4kW attenuator;

      (5) RTCU, and

      (6) fault locating indicator.

      (7) localizing fault to LRU;

(8) repairing and aligning system;

(9) verifying operational readiness; and

(10) powering down.

4. Time Required:

a. EO - 23 x 50 min periods; and

b. PC – 8 x 50 min periods.

5. Instructional Methods:

a. lecture;

b. practice; and

c. PC.

6. References:

a. A1 – A7;

b. A8; and

c. A39 – A56.

7. Training Aids:

a. HF Transmit System with system interfaces and/or MPT;

b. audio visual aids; and

c. handouts.

8. Test Details:

a. EO 003.04 will be tested as part of PC 003 IAW Annex K; and

b. PC 003 and debrief to be conducted IAW Annex K with a 60 min time limit.

9. Remarks:

a. MPT may be utilized;

b. a signal flow project may be assigned;

c. assignments are to be completed during non-instructional time; and

d. 8 practice periods are allotted for overall system faultfinding prior to PC 003.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Maintain the HF Transmit System by describing the sub-assembly functions of the following: a. band pass filter; b. OCU; c. TSM; d. fault finding indicator; e. RTCU; and f. 4 KW attenuator	L	7	
TP2	1. Localizing fault to LRU , to include: a. initialize the HF Transmit System; b. observe fault indicators; c. utilize trouble-shooting flow charts; d. utilize BITE; and e. identify faulted LRU.	P	16	

TP3	1. Repairing and aligning system, to include: a. replace faulted LRU; b. perform alignment as necessary; and c. verify DIP switch position.			
TP4	1. Verify operational readiness , to include: a. perform full system test; and b. power down HF Transmit System.			

PO 004 Maintain the LF/MF/HF Receive System

EO 004.01 Describe the LF/MF/HF Receive System

1. Performance. Describe the LF/MF/HF receive system.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) LF/MF/HF Receive System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall describe the LF/MF/HF receive system, IAW refs, by:

   a. system operation;

   b. system configurations;

   c. system functions;

   d. physical layout of equipment; and

   e. indicators and controls.

4. Time Required:

   a. EO - 6.5 x 50 min periods.

5. Instructional Methods:

   a. lecture;

   b. demonstration; and

   c. practice.

6. References:

   a. A1 – A7;

   b. A8; and

   c. A57 – A73.

7. Training Aids:

   a. a LF/MF/HF Receive System and/or MPT;

   b. audio visual aids; and

   c. handouts.

8. Test Details:

   a. EO 004.01 will be tested as a part of PC 004 IAW Annex K.

9. Remarks:

   a. projects and exercises may be assigned; and

   b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. The LF/MF/HF Receive System, to include: a. describe system operation to include; b. system operation; c. modes of operation; and d. capabilities and limitations.	L	0.5	
TP2	1. Describe system configuration, to include: a. system block diagram; and b. general signal flow.	L	0.5	
TP3	1. Describe system functions, to include: a. familiarize with receiver; b. the pre-selector; c. cross point matrix; and d. LF/MF/HF multi-coupler.	L P	2 2	
TP4	1. Describe the physical layout of equipment, to include: a. ships physical arrangement; b. cabinet layout; and	D	1	

	c. antenna location.			
TP5	1. Indicator and Controls, to include: a. publication layout; b. location of controls and indicators; and c. built in test equipment.	L	0.5	

PO 004 Maintain the LF/MF/HF Receive System

EO 004.02 Initialize the LF/MF/HF Receive System

1. Performance. Initialize the LF/MF/HF receive system

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) LF/MF/HF Receive System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize the LF/MF/HF receive system, IAW refs, by:

   a. powering up;

   b. carrying out equipment functional checks;

   c. verifying operational readiness;

   d. reporting equipment status to supervisor; and

   e. powering down.

4. Time Required:

   a. EO - 1x 50 min periods.

5. Instructional Methods:

   a. demonstration.

6. References:

   a. A1 – A7;

   b. A8; and

   c. A57 – A73.

7. Training Aids:

   a. LF/MF/HF Receive System with system interfaces and/or MPT;

   b. audio visual aids; and

   c. handouts.

8. Test Details:

   a. EO 004.02 will be tested as part of PC 004 IAW Annex K.

9. Remarks:

   a. MPT may be utilized;

   b. a signal flow project may be assigned; and

   c. assignments are to be completed during non-instructional time;

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize the LF/MF/HF Receive System, to include: a. verify that all controls and switches are in the correct position; b. perform Power up.	D	1	
TP2	1. Carry out equipment functional checks to, include: a. perform BITE tests; and b. interpret BITE test.			
TP3	1. Verify operational readiness by tuning to a known frequency (i.e. GMT clock).			
TP4	1. Report equipment status to supervisor.			

PO 004 Maintain the LF/MF/HF Receive System

EO 004.03 Maintain the LF/MF/HF Receive System

1. Performance. Maintain the LF/MF/HF receive system.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) LF/MF/HF Receive System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the LF/MF/HF Receive System, IAW refs, by:

   a. localizing fault to LRU;

   b. repairing and aligning system; and

   c. verifying operational readiness.

4. Time Required:

   a. EO - 8 x 50 min periods; and

   b. PC – 8 x 50 min periods.

5. Instructional Methods:

   a. practice; and

   b. PC.

6. References:

   a. A1 – A7;

   b. A8; and

   c. A57 – A73.

7. Training Aids:

   a. LF/MF/HF Receive System with system interfaces and/or MPT;

   b. audio visual aids;

   c. handouts; and

   d. references.

8. Test Details:

   a. EO 004.03 will be tested as part of PC 004 IAW Annex K; and

   b. PC 004 will be conducted IAW Annex K with a 45 min time limit.

9. Remarks:

   a. MPT may be utilized;

   b. a signal flow project may be assigned;

   c. assignments are to be completed during non-instructional time;

   d. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Maintain the LF/MF/HF Receive System, to include: a. localizing fault to LRU by; b. initialize the LF/MF/HF Receive System; c. observe fault indicators; d. utilize trouble-shooting flow charts; e. utilize BITE; and f. identify faulted LRU.	P	8	
TP2	1. Repairing and aligning system. to include: a. replace faulted LRU; and b. verify DIP switch position.			
TP3	1. Verify operational readiness, to include: a. tune to known frequency (i.e. GMT clock); and b. power down LF/MF/HF Receive System.			

PO 005 Maintain the Communications Control and Monitoring System

EO 005.01 Describe the Communications Control and Monitoring System

1. Performance. Describe Communications Control and Monitoring System (CCMS).

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CCMS; and

      (7) assistance..

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe Communications control / monitoring system, IAW refs, by:

   a. system operation;

   b. system configuration;

   c. system functions;

   d. physical layout of equipment; and

   e. indicators and controls.

4. Time Required:

a. EO-16 x 50 min periods.

5. Instructional Methods:

a. lecture;

b. demonstration; and

c. practice.

6. References:

a. A1 – A7;

b. A8; and

c. A74 – A88.

7. Training Aids.

a. communications control /monitoring Systems and/or MPT;

b. audio visual aids; and

c. handouts.

8. Test Details:

a. EO 005.01 will be tested as part of PC 005 IAW Annex K.

9. Remarks:

a. projects and exercises may be assigned; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Communications Control and Monitoring Systems, to include: a. describe System operation to include; b. system operation; c. modes of operation; and d. capabilities and limitations.	L	1	
TP2	1. Describe system configuration, to include: a. system block diagram; and b. general signal flow.	L	1	
TP3	1. Describe system functions, to include: a. TMS; b. CCMS; c. CCMS discrete IO; d. signal level converter; e. familiarize with WADS; and f. black patch panel (BPP); 2. primary and secondary back up modes, to include;	L P	7 4	(Practical for TMS job sheets)

	a. MMU; and b. CMTU.			
TP4	1. Describe Physical layout of equipment to include: a. ship physical arrangement; and b. cabinet layout.	D	2	
TP5	1. Indicators and controls to include: a. familiarize with publication layout; and b. familiarize with location of controls and indicators.	D	1	

PO 005 Maintain the Communications Control and Monitoring System

EO 005.02 Initialize the Communications Control and Monitoring System

1. Performance. Initialize the Communications control / monitoring System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CCMS; and

      (7) assistance..

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize the Communications control monitoring System, IAW refs, by:

   a. powering up in the correct sequence;

   b. loading and verifying software;

   c. verifying equipment functional checks;

   d. verifying indicator status;

   e. verifying system readiness;

   f. reporting equipment status to supervisor; and

   g. powering down in the correct sequence.

4. Time Required:

   a. EO - 6 x 50 min periods.

5. Instructional Methods.

   a. demonstration; and

   b. practice.

6. References:

   a. A1 – A7;

   b. A8; and

   c. A74 – A88.

7. Training Aids:

   a. Communications control / monitoring Systems and/or MPT;

   b. handouts; and

   c. audio visual aids.

8. Test Details:

   a. EO 005.02 will be tested as part of PC 005 IAW Annex K.

9. Remarks:

   a. MPT may be utilized,

   b. projects and exercises may be assigned; and

   c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize the Communications Control and Monitoring Systems, to include: a. verify that all controls and switches are in the correct position; b. perform Power up.			
TP2	1. Load and verify software by: a. load operational software; and b. verify successful load.			
TP3	1. Verify equipment functional checks by verifying status page.			
TP4	1. Verify Indicator Status by verifying alarm status.			
TP5	1. Verify system readiness at the DTS.			
TP6	1. Report equipment status to supervisor.			
TP7	1. Perform Power down in the correct sequence.			

PO 005 Maintain Communications Control and Monitoring System Software

EO 005.03 Initialize a Communication Circuit Utilizing the Black Patch Panel

1. Performance. Initialize a communication circuit utilizing the black patch panel.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CCMS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize a communication circuit, IAW refs, by:

   a. Configuring the black patch panel, to include:

      (1) plain voice circuit;

      (2) secure voice circuit; and

      (3) link data circuit.

4. Time Required:

   a. EO -28 x 50 min periods.

5. Instructional Methods:

a. lecture; and

b. practice.

6. References:

a. A1 – A7;

b. A8; and

c. A74 – A88.

7. Training Aids:

a. CCMS and/or MPT;

b. audio visual aids;

c. handouts; and

d. references.

8. Test Details:

a. EO 005.03 will be tested as part of PC 005 IAW Annex K.

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned; and

c. assignments are to be completed during non-instructional time;

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe black patch panel, to include: a. Patch panel layout; and b. Circuit requirements.	L	1	
TP2	1. Configure the black patch panel to include: a. Plain Voice Circuit; b. Secure Voice Circuit; and Link Circuit.	P	27	

PO 005 Maintain the Communications Control and Monitoring System

EO 005.04 Maintain the Communications Control and Monitoring System

1. Performance. Maintain the Command /Control Monitoring System (CCMS).

2. Conditions.

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CCMS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the CCMS, IAW refs, by:

   a. localizing fault to LRU;

   b. repairing and aligning system; and

   c. verifying system readiness.

4. Time Required.

   a. EO - 20 x 50 min periods; and

   b. PC – 8 x 50 min periods.

5. Instructional Methods.

a. practice; and

b. PC.

c. References.

d. A1 – A7;

e. A8; and

f. A74 – A88.

6. References:

a. A1 – A7;

b. A8; and

c. A74 – A88.

7. Training Aids.

a. CCMS with system interfaces and/or MPT;

b. audio visual aids; and

c. handouts;

8. Test Details.

a. EO 005.04 will be tested as part of PC 005 IAW Annex K; and

b. PC 005 and debrief to be conducted IAW Annex K with a 45 min time limit.

9. Remarks.

a. MPT may be utilized;

b. a signal flow project may be assigned;

c. assignments are to be completed during non-instructional time;

d. practice faultfinding on equipment will always be done in teams of 2;

e. 12 practice periods are allocated for software faultfinding;

f. 8 practice periods are allotted for overall system faultfinding prior to PC 005; and

g. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Localizing fault to LRU, to include: a. initialize the CCMS; b. observe fault indicators; c. perform/interpret BITE; and d. identify faulted LRU.	P	20	
TP2	1. Repairing and aligning system, to include: a. replace faulted LRU; b. verify DIP switch position; c. perform alignment as necessary; and d. perform/verify BITE as necessary.			
TP3	1. Verify system readiness, to include: a. verify alarm status; and b. power down CCMS in correct sequence.			

PO 006 Maintain the UHF Satellite Communications System

EO 006.01 Describe the UHF Satellite Communications System

1. Performance. Describe the UHF satellite communications system.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) UHF satellite communications system; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the UHF satellite communications system, IAW refs, by:

   a. system operation;

   b. system configurations;

   c. system functions;

   d. physical layout of equipment; and

   e. indicators and controls.

4. Time Required:

   a. EO – 7 x 50 min periods.

5. Instructional Methods:

   a. lecture; and

   b. demonstration.

6. References:

   a. A1 – A7;

   b. A8;

   c. A22 – A38;

   d. A89 – A96; and

   e. A97 – A102.

7. Training Aids:

   a. UHF satellite communication system and/or MPT;

   b. audio visual aids; and

   c. handouts.

8. Test Details:

   a. EO 006.01 will be tested as part of PC 006 IAW Annex K.

9. Remarks:

   a. projects and exercises may be assigned; and

   b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe system operation, to include; a. system operation; and b. capabilities and limitations.	L	1	
TP2	1. Describe system configuration, to include: a. system block diagram; and b. general signal flow.	L	1	
TP3	1. Describe system function, to include: a. OE-82C; and b. DAMA.	L	2	
TP4	1. Describe physical layout of equipment, to include: a. ship physical arrangement; b. cabinet layout; and c. antenna location.	D	2	
TP5	1. Indicators and Controls, to include: a. familiarize with publication layout;	L	1	

	b. familiarize with location of controls and indicators; and c. built in test equipment.			

PO 006 Maintain the UHF Satellite Communications System

EO 006.02 Initialize the UHF Satellite Communications System

1. Performance: Initialize the UHF satellite communication systems.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) UHF satellite communication system; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall initialize the UHF satellite communication systems, IAW refs, by:

   a. powering up;

   b. carrying out equipment functional checks;

   c. verifying operational readiness;

   d. reporting equipment status to supervisor; and

   e. powering down.

4. Time Required:

   a. EO - 4 x 50 min periods.

5. Instructional Methods:

   a. demonstration; and

   b. practice.

6. References:

   a. A1 – A7;

   b. A8;

   c. A22 – A38;

   d. A89 – A96; and

   e. A97 – A102.

7. Training Aids.

   a. UHF satellite communication systems and/or MPT;

   b. software;

   c. audio visual aids; and

   d. handouts.

8. Test Details:

   a. EO 006.02 will be tested as part of PC 006 IAW Annex K.

9. Remarks:

   a. MPT may be utilized;

   b. projects and exercises may be assigned; and

   c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Power up the UHF Satellite Communication System, to include: a. verify that all controls and switches are in the correct position; and b. perform power up.	D P	0.5 0.25	
TP2	1. Carry out equipment functional checks by performing and interpreting BITE tests.			
TP3	1. Verify operational readiness by checking status indicators.			
TP4	1. Report equipment status to supervisor.			
TP5	1. Perform power down.			

PO 006 Maintain the UHF Satellite Communications

EO 006.03 Maintain the OE-82C

1. Performance. Maintain the OE-82C.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) OE-82C and/or MPT

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the OE-82C, IAW refs, by:

   a. describing OE-82C;

   b. localizing fault to LRU;

   c. repairing and aligning system; and

   d. verifying operational readiness.

4. Time Required:

   a. EO - 15 x 50 min periods.

5. Instructional Methods:

a. lecture; and

b. practice.

6. References:

a. A1 – A7;

b. A8; and

c. A89 – A96.

7. Training Aids:

a. OE-82C and/or MPT;

b. software;

c. handouts;

d. audio visual aids; and

e. references.

8. Test Details:

a. EO 006.03 will be tested as part of PC 006 IAW Annex K.

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned; and

c. assignments are to be completed during non-instructional time;

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Maintain the OE-82C, to include: a. describe overall OE-82C function; b. reproduce block diagram for OE-82C; and c. describe controls and indicators for OE-82C. 2. Identify and describe sub-assemblies functions for OE-82C, to include: a. Antenna b. antenna controller; c. switching unit; and d. amplifier filter unit. 3. Describe capabilities and limitations of the equipment.	L	7	
TP2	1. Localizing fault to LRU, to include a. initialize the OE-82C; b. observe fault indicators; c. utilize trouble-shooting flow charts; and d. identify faulted LRU.	P	8	

TP3	1. Repair and align system by replacing faulted LRU			
TP4	1. Verify operational readiness , to include: a. manually rotate antennas; b. power down.			

PO 006 Maintain the UHF Satellite Communications

EO 006.04 Maintain the Demand Assign Multiple Access System

1. Performance. Maintain the Demand Assign Multiple Access System (DAMA).

2. Conditions:

   a. Given

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) DAMA and/or MPT

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain DAMA, IAW refs, by:

   a. describing DAMA;

   b. localizing fault to LRU;

   c. repairing and aligning system; and

   d. verifying operational readiness.

4. Time Required:

   a. EO - 20 x 50 min periods.

   b. PC – 8 x 50 min periods.

5. Instructional Methods:

   a. lecture; and

   b. practice.

6. References:

   a. A1 – A7;

   b. A8; and

   c. A97 – A102.

7. Training Aids:

   a. DAMA and/or MPT;

   b. software;

   c. handouts; and

   d. audio visual aids.

8. Test Details:

   a. EO 006.04 will be tested as part of PC 006 IAW Annex K.

9. Remarks:

   a. MPT may be utilized;

   b. projects and exercises may be assigned; and

   c. assignments are to be completed during non-instructional time;

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe DAMA, to include: a. describe overall DAMA function: b. reproduce block diagram for DAMA; c. describe controls and indicators for DAMA; and d. identify and describe sub-assemblies functions for DAMA.; 1) switchboard; and 2) crypto unit. e. Describe capabilities and limitations of the equipment.	L	12	
TP2	1. Localizing fault to LRU by: a. initialize the DAMA; b. observe fault indicators; c. utilize trouble-shooting flow charts; and d. identify faulted LRU.	P	8	
TP3	1. Repair and align system, to include:			

	a. replace faulted LRU.			
TP4	1. Verify operational readiness , to include: a. verify control indicators; and b. power down.			

PO 007 Maintain Satellite Communication System

EO 007.01 Describe the Fleet Broadband (FBB)

1. Performance. Describe the FBB.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communication System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the FBB, IAE refs, by:

   a. Describing the FBB to include:

   b. system operation;

   c. system configurations;

   d. system functions;

   e. physical layout of equipment; and

   f. indicators and controls.

   g. Manipulate the FBB software through the handset.

4. Time Required:

   a. EO – 14 x 50 min periods.

5. Instructional Methods:

   a. lecture;

   b. demonstration; and

   c. practice.

6. References:

   a. A1 – A7; and

   b. A103 – A111

7. Training Aids:

   a. FBB;

   b. audio visual aids; and

   c. handouts.

8. Test Details:

   a. EO 007.01 will be tested in EC 007.01 and PC 007.

9. Remarks:

   a. projects and exercises may be assigned; and

   b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe the FBB operation, to include: a. system operation including spot beam selection; and b. capabilities and limitations to include enhanced-fit operation.	L	1	
TP2	1. Describe system configuration, to include: a. block diagram; and b. general signal flow.	L	1	
TP3	1. Describe system function, to include: a. antenna; b. main control unit; c. handset controller and software	L	2	
TP4	1. Describe Physical layout of equipment, to include: a. ship physical arrangement; b. location of controls and indicators; and c. built in test equipment.	D	1	

TP 5	1. Indicators and controls: a. familiarize with publication layout; b. familiarize with location of controls and indicators; and c. describe Built in test equipment.	L	1	
TP6	1. Manipulate the FBB software.	P	8	

PO 007 Maintain Satellite Communication System

EO 007.02 Initialize the FBB

1. Performance. Initialize the FBB

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize the FBB system, IAW refs, by:

   a. powering up;

   b. carrying out equipment functional checks;

   c. verifying operational readiness;

   d. reporting equipment status to supervisor; and

   e. powering down.

4. Time Required:

   a. EO – 1 x 50 min periods.

5. Instructional Methods:

   a. demonstration.

6. References:

   a. A1 – A7; and

   b. A103 – A111

7. Training Aids:

   a. FBB;

   b. software;

   c. audio visual aids; and

   d. handouts.

8. Test Details:

   a. EO 007.02 will be tested as part of PC 007IAW Annex K.

9. Remarks:

   a. projects and exercises may be assigned; and

   b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize the FBB, to include: a. Power up to include; and b. verify that all controls and switches are in the correct position.	D	1	
TP2	1. Carry out equipment functional checks by performing and interpreting BITE tests.			
TP3	1. Verify operational readiness by checking status indicators.			
TP4	1. Report equipment status to supervisor.			
TP5	1. Perform power down.			

PO 007 Maintain Satellite Communication System

EO 007.03 Maintain the FBB

1. Performance. Maintain the FBB.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the FBB system, IAW refs, by:

   a. localizing fault to LRU;

   b. repairing and aligning system; and

   c. verifying operational readiness.

4. Time Required:

   a. EO – 8 x 50 min periods; and

   b. PC – 8 x 50 min periods.

5. Instructional Methods:

a. practice; and

b. PC.

6. References:

a. A1 – A7; and

b. A103 – A111

7. Training Aids:

a. FBB;

b. software;

c. handouts; and

d. audio visual aids.

8. Test Details:

a. EO 007.03 will be tested in PC 007, IAW Annex K; and

b. PC 007 and debrief to be conducted IAW Annex K with a 45 min time limit.

9. Remarks:

a. projects and exercises may be assigned;

b. assignments are to be completed during non-instructional time; and

c. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Localizing fault to LRU, to include: a. initialize the FBB; b. connect to satellite; c. select modes of operation; d. observe fault indicators; e. utilize trouble-shooting flow charts; and f. identify faulted LRU.	P	8	
TP2	1. Repairing and aligning system by replacing faulted LRU.			
TP3	1. Verify operational readiness, to include: a. verify control indicators; b. make local call; and c. power down.			

PO 007 Maintain Satellite Communications System

EO 007.04 Describe the Maritime Satellite Communications Unit

1. Performance: Describe the Maritime Satellite Communications Unit

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall describe the MSCU system, IAW refs, by:

   a. Describing the Basic and Fundamental Satellite Communications;

   b. Describing the Satellite Communications to include:

   c. General, Physical, Functional, Operational and Interface Characteristics of the terminal;

   d. Terminal Controls and Indicators;

   e. Configure and Operate the Terminal for Mission Performance

   f. Safety Precautions Concerning Operation and Maintenance of the Terminal; and

g. Terminal Circuits and Signal Flow.

4. Time Required:

a. EO – 21 x 50 min periods.

5. Instructional Methods:

a. Lecture; and

b. Demonstration.

6. References:

a. A1 - A26, A27, A153, C1 – C2.

7. Training Aids:

a. handouts; and

b. audio visual aids.

8. Test Details:

a. EO 008.01 will be tested in PC 008, IAW Annex K; and

b. PC 008 and debrief to be conducted IAW Annex K with a 45 min time limit.

9. Remarks:

a. Projects and exercises may be assigned;

b. Assignments are to be completed during non-instructional time;

c. practice faultfinding on equipment will always be done in teams of 2; and

d. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Instructional Method	Time Required	Reference
TP1	1. Basic and Fundamental Satellite Communication: a. Satellite orbital Operation b. Satellite signal polarization standards	L	1	
TP2	1. General, physical, functional, operational, and interface characteristics of the terminal to include: a. Major Components b. Transmit / Receive Signal Flow c. 10MHz Signal Flow 2. Power Requirements	L D	5	
TP3	1. Locate, describe, and manipulate the MSCU terminal: a. Identify below deck equipment lights and indicators b. Manipulate MSCU below deck equipment	L D/P	3	
TP4	1. Manipulate the OI software:	L	5.5	

	a. Manipulate the Terminal Configuration Window b. Manipulate the Online Equipment Window c. Manipulate Antenna Control Window d. Manipulate Alarm/Event Window e. Manipulate the Configuration Tab f. Manipulate Status Tab ~~a.~~g. Manipulate Alarm Tab	D/P		
TP5	1. Describe Safety Precautions concerning operation and maintenance of the terminal a. Describe MAN-A-Loft procedure b. Describe EMCON procedure c. Describe safety procedures for CM	L	1.5	
TP 6	1. Review and Understand Terminal Circuits and Signal Flow at a detailed block diagram level a. Describe general, physical, functional, operational, and interface characteristics of the systems components b. Describe transmit and receive signal flow	L D	5	

	c. Describe 10MHz signal flow d. Describe system power requirements			

PO 007 Maintain Satellite Communication System

EO 007.05 Initialize the Maritime Satellite Communications Unit

1. Performance: Initialize the Maritime Satellite Communications Unit

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (2) all shipboard conditions at sea or alongside; and

      (3) day or night.

3. Standard: The Trainee shall Initialize the Maritime Satellite Communications Unit, IAW refs, by:

   a. Powering Up;

   b. Carrying out equipment functional checks;

   c. Verifying operational readiness

   d. Reporting equipment Status to Supervisor; and

   e. Powering Down.

4. Time Required:

   a. EO – 5.5 x 50 min periods.

5. Instructional Methods:

   a. practice; and

   b. PC.

6. References:

   a. A1 - A26, A27, A153, C1 – C2.

7. Training Aids:

   a. Handouts; and

   b. Audio visual aides.

8. Test Details:

   a. EO 008.02 will be tested in PC 008, IAW Annex K; and

   b. PC 008 and debrief to be conducted IAW Annex K with a 45 min time limit.

9. Remarks:

   a. projects and exercises may be assigned;

   b. assignments are to be completed during non-instructional time;

   c. practice faultfinding on equipment will always be done in teams of 2; and

   d. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Conduct Terminal Startup Procedures a. Conduct proper flash-up routine b. Identify and correct any minor faults within the BDE cabinet and remote OIU	L D P	4.5	
TP2	1. Conduct Shutdown Routine	P	1	

PO 007 Maintain Satellite Communications System

EO 007.06 Maintain the Maritime Satellite Communications Unit

1. Performance: Maintain the Maritime Satellite Communications Unit.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall maintain the MSCU system, IAW refs, by:

   a. Localizing fault to LRU;

   b. Repairing and aligning system; and

   c. Verifying operational readiness.

4. Time Required:

   a. EO – 39.5 x 50 min periods; and

   b. PC – 8 x 50 min periods.

5. Instructional Methods:

a. practice; and

b. PC.

6. References:

a. A1 - A26, A27, A153, C1 – C2.

7. Training Aids:

a. handouts; and

b. audio visual aids.

8. Test Details:

a. EO 008.03 will be tested in PC 008, IAW Annex K; and

b. PC 008 and debrief to be conducted IAW Annex K with a 45 min time limit.

9. Remarks:

a. projects and exercises may be assigned;

b. assignments are to be completed during non-instructional time;

c. practice faultfinding on equipment will always be done in teams of 2; and

d. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Perform test and inspections to verify system operability a. Perform available online and offline tests b. Verify system is operational	P	6	
TP2	1. Demonstrate the ability to troubleshoot/fault isolate equipment performance issues on the Terminal a. Alarm/Event Window b. Online Equipment Window c. Antenna Control Window d. All other fault indicators on below-deck equipment e. Isolate issues to faulted components	L D/P	12.5	
TP3	1. Perform general routine and periodic	L/D	4	

	preventative maintenance actions a. Conduct proper safety routines for maintenance b. Carrying out maintenance routine c. Returning system to service	P		
TP4	1. Demonstrate the ability of how to remove and replace faulted LRU components on both the below deck and above deck terminal components a. Conduct component removal/replacement of the above-deck terminal components to include: 1) Feed-horn 2) High Powered Amplifier (HPA) b. Conduct component removal/replacement of below-deck terminal components to include: 1) 10MHZ reference signal bypass	P	17	

	2) Internal network component			

PO 007 Maintain Satellite Communications System

EO 007.07 Describe the Naval Multiband Terminal

1. Performance: Describe the Naval Multiband Terminal

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall Describe the NMT system, IAW refs, by:

   a. Describing the Basic and Fundamental Satellite Communications;

   b. Describing the Satellite Communications to include:

   c. General, Physical, Functional, Operational and Interface Characteristics of the terminal;

   d. Terminal Controls and Indicators;

   e. Configure and Operate the Terminal for Mission Performance

   f. Safety Precautions Concerning Operation and Maintenance of the Terminal; and

g. Terminal Circuits and Signal Flow.

4. Time Required:

a. EO – 21 x 50 min periods.

5. Instructional Methods:

a. Lecture; and

b. Demonstration.

6. References:

a. A1 - A26, A27, A153, C1 – C2.

7. Training Aids:

a. handouts; and

b. audio visual aids.

8. Test Details:

a. EO 007.07 will be tested in PC 007, IAW Annex K; and

b. PC 007 and debrief to be conducted IAW Annex K with a 45 min time limit.

9. Remarks:

a. Projects and exercises may be assigned;

b. Assignments are to be completed during non-instructional time;

c. practice faultfinding on equipment will always be done in teams of 2; and

d. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Basic and Fundamental Satellite Communication: a. Satellite orbital Operation b. Satellite signal polarization standards	L	1	
TP2	1. General, physical, functional, operational, and interface characteristics of the terminal to include: a. Major Components b. Transmit / Receive Signal Flow c. Power Requirements	L D	5	
TP3	1. Locate, describe, and manipulate the NMT terminal: a. Identify below deck equipment lights and indicators b. Manipulate NMT below deck equipment	L D/P	3	

TP4	1. Manipulate the OIU software: a. Manipulate the Terminal Configuration Window b. Manipulate the Online Equipment Window c. Manipulate Antenna Control Window d. Manipulate Alarm/Event Window e. Manipulate the Configuration Tab f. Manipulate Status Tab g. Manipulate Alarm Tab	L D/P	5.5	
TP5	1. Describe Safety Precautions concerning operation and maintenance of the terminal a. Describe MAN-A-Loft procedure b. Describe EMCON procedure c. Describe safety procedures for CM	L	1.5	
TP6	1. Review and Understand Terminal Circuits and Signal Flow at a detailed block diagram level a. Describe general, physical, functional, operational, and interface characteristics	L D	5	

	of the systems components b. Describe transmit and receive signal flow c. Describe system power requirements			

PO 007 Maintain Satellite Communication System

EO 007.08 Initialize the Naval Multiband Terminal

1. Performance: Initialize the Naval Multiband Terminal

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall Initialize the Naval Multiband Terminal, IAW refs, by:

   a. Powering Up;

   b. Carrying out equipment functional checks;

   c. Verifying operational readiness

   d. Reporting equipment Status to Supervisor; and

   e. Powering Down.

4. Time Required:

   a. EO – 5.5 x 50 min periods.

5. Instructional Methods:

   a. practice; and

   b. PC.

6. References:

   a. A1 - A26, A27, A153, C1 – C2.

7. Training Aids:

   a. Handouts; and

   b. Audio visual aides.

8. Test Details:

   a. EO 007.08 will be tested in PC 007, IAW Annex K; and

   b. PC 007 and debrief to be conducted IAW Annex K with a 45 min time limit.

9. Remarks:

   a. projects and exercises may be assigned;

   b. assignments are to be completed during non-instructional time;

   c. practice faultfinding on equipment will always be done in teams of 2; and

   d. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP7	1. Conduct Terminal Startup Procedures a. Conduct proper flash-up routine b. Identify and correct any minor faults within the cabinet and remote OIU	L D P	4.5	
TP8	1. Conduct Shutdown Routine	P	1	

PO 007 Maintain Satellite Communications System

EO 007.09 Maintain the Naval Multiband Terminal

1. Performance: Maintain the Naval Multiband Terminal.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Satellite Communications System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) all shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall maintain the NMT system, IAW refs, by:

   a. Localizing fault to LRU;

   b. Repairing and aligning system; and

   c. Verifying operational readiness.

4. Time Required:

   a. EO – 39.5 x 50 min periods; and

   b. PC – 8 x 50 min periods.

5. Instructional Methods:

a. practice; and

b. PC.

6. References:

a. A1 - A26, A27, A153, C1 – C2.

7. Training Aids:

a. handouts; and

b. audio visual aids.

8. Test Details:

a. EO 007.09 will be tested in PC 007, IAW Annex K; and

b. PC 007 and debrief to be conducted IAW Annex K with a 45 min time limit.

9. Remarks:

a. projects and exercises may be assigned;

b. assignments are to be completed during non-instructional time;

c. practice faultfinding on equipment will always be done in teams of 2; and

d. PC is 1 period (45 min for PC and 5 min for debrief) corrective measures start at minute 45 if extra time is required.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Perform test and inspections to verify system operability a. Perform available online and offline tests b. Verify system is operational	P	6	
TP2	1. 1. Demonstrate the ability to troubleshoot/fault isolate equipment performance issues on the Terminal a. Alarm/Event Window b. Online Equipment Window c. Antenna Control Window d. All other fault indicators on below-deck equipment e. Isolate issues to faulted components	L D/P	12.5	
TP3	1. Perform general routine and periodic	L/D	4	

	preventative maintenance actions a. Conduct proper safety routines for maintenance b. Carrying out maintenance routine c. Returning system to service	P		
TP4	1. Demonstrate the ability of how to remove and replace faulted LRU components on both the below deck and above deck terminal components a. Conduct component removal/replacement of the above-deck terminal components. b. Conduct component removal/replacement of below-deck terminal components.	P	17	
TP5	1. Verify operational readiness by acquiring the satellite	L/D P	1	

PO 008 **Not Allocated**

PO 009 Maintain Tactical Data Systems

EO 009.01 Describe the Link Systems

1. Performance. Describe the Link systems.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Multi-Link and peripherals

      (7) software and crypto if needed; and

      (8) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the Link systems, IAW refs, by:

   a. system operation;

   b. system configuration;

   c. system functions;

   d. physical layout of equipment; and

   e. indicators and controls.

4. Time Required:

a. EO - 9 x 50 min periods.

5. Instructional Methods:

a. lecture; and

b. demonstration.

6. References:

a. A1 – A7;

b. A8;

c. A112 – A115; and

d. C5.

7. Training Aids:

a. Link system and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 009.01 will be tested with PC 009 IAW Annex K.

9. Remarks:

a. projects and exercises may be assigned; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. System Operation to include: a. describe system operation; b. describe modes of operation; c. capabilities and limitations;	L	1	
TP2	1. System configurations, to include: a. reproduce block diagram to include connectivity to CMS; and b. describe general signal flow to include connectivity to communication systems	L	1	
TP3	1. System functions, to include: a. Link 16 MIDS/JTRS; b. Link 22 SPU, SNC; and c. JREAP – C.	L	5	

TP4.	1. Physical layout of equipment to include: a. describe ship physical arrangement; and b. describe cabinet layout.	D	1	
TP5.	1. Indicators and controls, to include: a. familiarize with publication layout; b. familiarize with location of controls and indicators; and c. describe built in test equipment (BIT).	L	1	

PO 009 Maintain Tactical Data Systems

EO 009.02 Initialize the Link Systems

1. Performance. Initialize the Link systems.

2. Conditions:

   a. Given

      (1) supervision

      (2) Link System and/or MPT;

      (3) references;

      (4) DWAN;

      (5) tools and test equipment;

      (6) PPE;

      (7) Link system and peripherals

      (8) software and crypto if needed; and

      (9) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall initialize the Link 11 System, IAW refs, by:

   a. power up;

   b. carry out equipment functional checks ;

   c. verifying software load;

   d. verifying operational readiness;

   e. reporting equipment status to supervisor; and

f. powering down.

4. Time Required:

a. EO – 3 x 50 min periods.

5. Instructional Methods:

a. lecture ;

b. demonstration; and

c. practical.

6. References:

a. A1 – A7;

b. A8;

c. A112 – A115; and

d. C5.

7. Training Aids:

a. MPT or a Link system;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 009.02 will be tested in PC 009, IAW Annex K.

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize the Link systems, to include: a. Perform Power-Up; and b. Verify Operational Status.	D P	1 2	
TP2	1. Verify operational readiness by checking Link Status in CMS.			
TP3	1. Report equipment status to supervisor.			
TP4	1. Perform Power down.			

PO 009 Maintain Tactical Data System

EO 009.03 Repair the Link System

1. Performance. Repair the Link system.

2. Conditions:

   a. Given

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Multi-Link and peripherals

      (7) software and crypto if needed; and

      (8) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall repair the Link System, IAW refs, by:

   a. Initializing the system;

   b. localize fault to LRU;

   c. repair and align system;

   d. verify operational readiness; and

   e. power down.

4. Time Required:

a. EO – 8 x 50 min periods; and

b. PC – 8 X 50 min periods.

5. Instructional Methods:

a. practical.

6. References:

a. A1 – A7;

b. A8;

c. A112 – A115;

d. C5.

7. Training Aids:

a. MPT or a Link system;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 009.03 will be tested as part of PC 009; and

b. PC 009 and debrief will be conducted with a 50 min time limit (Corrective measures start at min 45).

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned;

c. assignments are to be completed during non-instructional time; and

d. practice faultfinding on equipment will always be done in teams of 2.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Localizing fault to LRU, to include: a. initialize the Link System; b. observe fault indicators; c. utilize trouble-shooting flow charts; and d. identify faulted LRU.			
TP2	1. Repairing system by replacing faulted LRU.			
TP3	1. Verifying operational readiness, to include: a. verify control indicators; b. check connectivity using software function; and c. Power down.			

PO 009 Maintain Tactical Data System

EO 009.04 Describe the Data Link Processor System

1. Performance: Describe the data link processor system (DLPS).

2. Conditions:

   a. Given

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Multi-Link and peripherals

      (7) software and crypto if needed; and

      (8) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the DLPS, IAW refs, by:

   a. personal and equipment and Safety;

   b. publication layout;

   c. Multi-LINK System block diagram;

   d. capabilities and limitations;

   e. theory of operations;

   f. power distribution; and

g. control and indicators of major components.

4. Time Required:

a. EO –10 x 50 min periods; and

b. EC – 1 x 50 min periods.

5. Instructional Methods:

a. lecture ; and

b. assignments.

6. References:

a. A1 – A7;

b. A8; and

c. A116 – A119.

7. Training Aids:

a. MPT or a Multi-Link system;

b. audio visual aids; and

c. handouts.

8. Test Details:

a. EO 009.04 will be tested in EC 009.04 and PC 009-2.

9. Remarks:

a. MPT may be utilized,

b. projects and exercises may be assigned; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe the Multi-Link system, to include: a. Describe system operation to, include; 1) DLPS Architecture; 2) TOD Interface Manager; and 3) Circuit Cards.	L P	1	
TP2	1. Capabilities and Limitations, to include: a. DLPS; b. Link systems; and c. JREAP.	L P	1	
TP3	1. Theory of operation, to include; a. MULTILINK Super Networks; b. LINK 16; c. LINK 22; and d. JREAP.	L P	1	
TP4	1. Power Distribution, to include:	L P	1	

	a. power and voltage requirements; and b. breakers and switchboard connection.			
TP5	1. Controls and Indicators.	L P	1 1	

PO 009 Maintain Tactical Data Systems

EO 009.05 Initialize the Data Link Processor System

1. Performance: Initialize the data link processor system (DLPS).

2. Conditions:

   a. Given

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Multi-Link and peripherals

      (7) software and crypto if needed; and

      (8) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall initialize the DLPS,IAW refs, by:

   a. power up;

   b. carry out equipment functional checks ;

   c. verifying software load;

   d. verifying operational readiness;

   e. reporting equipment status to supervisor; and

   f. powering down.

4. Time Required:

   a. EO –10 x 50 min periods; and

   b. EC – 1 x 50 min periods.

5. Instructional Methods:

   a. lecture; and

   b. assignments.

6. References:

   a. A1 –A7;

   b. A8; and

   c. A116 – A119.

7. Training Aids:

   a. MPT or a Multi-Link system;

   b. references;

   c. audio Visual Aids; and

   d. handouts.

8. Test Details:

   a. EO 009.05 will be tested in EC 009.05 and PC 009-2.

9. Remarks:

   a. MPT may be utilized;

   b. projects and exercises may be assigned; and

   c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize the Multi-Link System, to include: a. system operation, to include; b. pre-operation procedure; c. flash-Up procedure; and d. shut Down procedure	L P	1	
TP2	1. Demonstrate/practice operating special test equipment, to include: a. maintainer terminal.	L P	1	
TP3	1. Initialize the Multi-LINK DLPS, to include:; a. switch-on procedures; b. verify System operation: and c. shut Down procedures.	L P	1	

TP4	1. Initialize the Multi-LINK DLPS, to include; a. pre-operation procedure; b. flash-up warnings; c. flash-up procedure; d. shut down procedure; and e. post operation procedure.	L P	1	

PO 009 Maintain Tactical Data Systems

EO 009.06 Repair the Data Link Processor System

1. Performance. Repair the data link processor system (DLPS)

2. Conditions:

   b. Given:

      (3) supervision;

      (4) references;

      (5) DWAN;

      (6) tools and test equipment;

      (7) PPE;

      (8) Multi-Link and peripherals

      (9) software and crypto if needed; and

      (10) assistance.

   c. Denied:

      (1) Nil.

   d. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall repair the DLPS, IAW refs, by:

   a. localizing faults to the DLPS;

   b. removal and replacement of LRUs; and

   c. verifying system operational readiness.

4. Time Required:

   a. EO – 16 x 50 min periods; and

   b. EC – 1 x 50 min periods.

c. PC - 8 X 50 min periods.

5. Instructional Methods:

a. lecture; and

b. MPT.

6. References:

a. A1 – A7;

b. A8; and

c. A116 –A119.

7. Training Aids:

a. MPT or a Multi-Link system;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 009.06 will be tested in EC 009.06 and PC 009-2.

b. PC 009.02 and debrief to be conducted with a 50 min time limit with corrective measures starting at min 45.

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Repair the DLPS: a. Major components of the DLPS, to include: 1) Localize faults to the DLPS; 2) removal and replacement of LRUs; and 3) methods of verifying system operational readiness.	L	2	
TP2	1. Repair the DLPS, to include; a. explain the major components of the DLPS; b. methods to localize faults to the DLPS; c. removal and replacement of LRUs; and d. verifying system operational readiness.	L P	1 9	
TP3	1. Review of MPT faults, to include: a. review faults in order.	L	2	

TP4	1. Review of MPT fault, to include: a. review of fault.	L P	1 1	

PO 009 Maintain Tactical Data Systems

EO 009.07 Conduct Preventative Maintenance for the Data Link Processor System

1. Performance. Conduct preventative maintenance for the data link processor system (DLPS)

2. Conditions:

   a. Given

      (3) supervision;

      (4) references;

      (5) DWAN;

      (6) tools and test equipment;

      (7) PPE;

      (8) Multi-Link and peripherals

      (9) software and crypto if needed; and

      (10) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard: The Trainee shall conduct PM for the DLPS, IAW refs, by;

   a. reasons for PM routines; and

   b. safety considerations;

   c. various types of routines; and

   d. content and use of PM routines

4. Time Required.

   a. EO – 2 x 50 min periods;

b. EC – 1 x 50 min periods; and

5. Instructional Methods:

a. lecture; and

b. MPT.

6. References:

a. A1 – A7;

b. A8; and

c. A116 – A119.

7. Training Aids:

a. MPT or a Multi-Link system;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 009.06 will be tested in EC 009.06 and PC 009-2 IAW Annex – K; and

9. Remarks:

a. MPT may be utilized;

b. projects and exercises may be assigned; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Preventative Maintenance for the DLPS, to include: a. reasons for PM routines; b. safety consideration; c. various types of routines; and d. content and use of PM routines.	L	2	

PO 010 Maintain Combat Management System 330

EO 010.01 Describe the Combat Management System 330

1. Performance. Describe the combat management system (CMS 330).

2. Conditions:

   b. Given:

      (3) CCMS and/or MPT;

      (4) tools and test equipment;

      (5) replacement parts;

      (6) references;

      (7) software;

      (8) supervision; and

      (9) assistance. as required.

   c. Denied:

      (1) Nil.

   d. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the CMS 330, IAW refs, by:

   a. system components;

   b. capabilities and limitations;

   c. theory of operation;

   d. initial power distribution; and

   e. indicators and controls.

4. Time Required.

   a. EO – 10 x 50 min periods;

5. Instructional Methods:

   a. lecture.

6. References:

   a. A1 – A7; and

   b. A116 – A119.

7. Training Aids:

   a. Combat Management System and/or MPT;

   b. audio visual aids;

   c. references; and

   d. handouts.

8. Test Details:

   a. EO 010.01 will be tested with PC 010.

9. Remarks:

   a. For the purpose of this EO, the student is assumed to have a working knowledge of CMS operation, initialization and maintenance. This training should be focused on the higher functions of CMS.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Describe system components ,to include; a. mission critical enclosures; b. multi-function workstation; and, c. local area network	L	2	
TP2	1. Capabilities and limitations to include; a. Cabinet enclosures; b. remote access server; c. cabinet cooling system; and d. MCE cabinets.	L	2	

TP3	1. Theory of operation ,to include; a. layered system of sensors; b. information exchange systems(IES); c. tracking tools; and d. decision aids and weapons to 1) detect 2) locate; 3) track; 4) identify; 5) assess; and 6) engage threats.	L	2	
TP4	1. Power Distribution, to include; a. Breaker and switchboard connection; b. MCE power; c. MFW power; and d. subassembly voltage input	L	2	

TP5	1. Indicators and controls, to include; a. Common MCE Components; b. MCE 1 and 2 Subassemblies; c. MCE 3 and 4 Subassemblies; d. MCE 5 Subassemblies; e. MCE 6 Subassemblies; f. MCE 7 Subassemblies; g. MCE 8 Subassemblies; h. MCE 11 Subassemblies; and i. MCE 12 Subassemblies	L	2	

PO 010 Maintain Combat Management System

EO 010.02 Initialize the Combat Management System

1. Performance. Initialize the combat management system (CMS).

2. Conditions:

   a. Given:

      (1) CMS and/or MPT;

      (2) tools and test equipment;

      (3) replacement parts;

      (4) references;

      (5) software;

      (6) supervision; and

      (7) assistance. as required.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize the CMS, IAW refs, by:

   a. powering up;

   b. conducting equipment functional checks;

   c. loading CMS software;

   d. verify indicator status;

   e. report equipment status to supervisor; and

   f. powering down.

4. Time Required.

a. EO – 10 x 50 min periods.

5. Instructional Methods.

a. lecture

b. demo; and

c. PC.

6. References:

a. A1 – A7; and

b. A116 – A119.

7. Training Aids:

a. a CMS 330 and/or MPT;

b. audio visual aids;

c. references; and

d. handouts.

8. Test Details:

a. EO 010.02 will be tested with PC 010 IAW Annex - K.

9. Remarks:

a. Practice by Flash-up and faultfinding will be conducted in teams of 2; and

b. EC/PC checks will be carried out individually.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Initialize CMS, to include; a. preliminary checks on associated equipment; b. power up; c. checks during and after power on; d. load and verify software; e. conduct system functional checks; f. verify operational readiness; g. report system status to supervisor; and h. power down.	D P	2 8	

PO 010 Maintain Combat Management System

EO 010.03 Repair the Combat Management System

1. Performance. Repair the combat management system (CMS)

2. Conditions:

   a. Given:

      (1) CMS and/or MPT;

      (2) tools and test equipment;

      (3) replacement parts;

      (4) references;

      (5) software;

      (6) supervision; and

      (7) assistance. as required.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall repair the CMS, IAW refs, by:

   a. powering up;

   b. localize the fault to the LRU;

   c. repair;

   d. verify operational readiness;

   e. report equipment status to supervisor; and

   f. powering down.

4. Time Required.

a. EO – 10 x 50 min periods; and

b. PC - 8 X 50 min periods.

5. Instructional Methods.

a. lecture

b. demo; and

c. PC.

6. References:

a. A1 – A7; and

b. A116 – A119.

7. Training Aids:

a. a Combat Management System and/or MPT;

b. audio visual aids;

c. references; and

d. handouts.

8. Test Details:

a. EO 007.02 will be tested with PC 007 IAW Annex K.

9. Remarks:

a. practice Flash up and faultfinding will be conducted in teams of 2; and

b. EC/PC checks will be carried out individually.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice.

	Description	Instructional Method	Time Required	Reference
TP1	1. Maintain CMS, by; a. symptom recognition; b. localize fault to LRU c. use of fault finding logic charts; d. logical procedure; e. interpretation of schematics; f. verify operational readiness; g. tests and alignments; h. describe removal and replacement of LRU's; and, i. verify operational readiness	P	10	

PO 011 Utilize DRMIS Portal as a Maintenance Supervisor

EO 011.01,.02,.03 IAW PO 002 – Utilize DRMIS Portal as a Maintenance Supervisor IAW QSP119097

1. Conditions:

   a. Given:

      (1) References;

      (2) Computer with access to DRMIS simulator; and

      (3) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) classroom

2. Standard: IAW PO 002 – Utilize DRMIS Portal as a Maintenance Supervisor (Mod 2) QSP119097.

3. Time Required:

   a. 5 x 50 min periods.

4. Instructional Methods:

   a. IAW PO 002 DRMIS QSP.

5. References:

   a. IAW PO 002 DRMIS QSP.

6. Training Aids:

   a. IAW PO 002 DRMIS QSP.

7. Test Details:

   a. IAW PO 002 DRMIS QSP.

8. Remarks:

   a. All Trainees require a level 1 security clearance.

9. Teaching Points:

a. IAW PO 002 DRMIS QSP.

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

REF CODE		PUBLICATION
		COMMON REFERENCES
A1.		Class Specific Safety and Environment Management System (SEMS)
A2.		B-GN-007-RCN/FP-002 CFCD 133 RCN ERT Manual
A3.		C-01-000-000/DA-001 NDID Coding System
A4.		Class Specific Ship's Standing Orders (SSOs)
A5.		CF H Svcs Gp Instruction 4440-01 Chemical Hazards Surveillance Program
A6.		Class Specific ERT Pack

A7.		DAOD 4003-0, Environmental Protection and Stewardship
		External Communications
A8.		C-51-501-000/MS-001 External Naval Communications System
		SHINCOM
A9.		C-52-476-000/MS-001 TECHNICAL MANUAL CPF SHINCOM 3100
A10.		C-52-476-000/MB-004 Operator's Guide Recorder Storage Unit Single Interface
A11.		C-52-476-000/MB-007 SHINCOM 3100-Technical Manual Automatic Software Upgrade (ASU) Console Application
A12.		C-52-476-000/MB-003 Operator's Guide Helicopter Audio Distribution System Single Interface Node
A13.		C-52-289-001/MY-001 Shipboard Integrated Interior Communications System AN/SSC-505(V), Parts Identification List
A14.		C-52-476-000/MB-001 OEPRATOR'S GUIDE SHINCOM 3100 DUAL SCREEN TERMINAL
A15.		C-52-476-000/MB-002 Outdoor Terminal Operator’s Guide

A16.		C-52-476-000/MB-005 SHINCOM 3100 – PUBLIC ADDRESS SINGLE INTERFACE NODE OPERATOR'S GUIDE
A17.		C-52-476-000/MB-006 Operator's Guide Console Dual Screen Terminal
A18.		C-52-476-000/MS-002 TECHNICAL MANUAL CONSOLE DUAL SCREEN TERMINAL
A19.		C-52-476-000/NK-001 NAVAL PREVENTIVE MAINTENANCE PERFORMANCE TEST COMMUNICATIONS) INTERIOR COMMUNICATIONS SYSTEM (SHINCOM) 3100
A20.		C-52-476-000/NT-001 READINESS TRIAL FOR INTERIOR COMMUNICATIONS SYSTEM (SHINCOM) 3100
A21.		C-52-476-000/NY-001 NAVAL PREVENTIVE MAINTENANCE SCHEDULE (COMMUNICATIONS) INTERIOR COMMUNICATIONS SYSTEM (SHINCOM) 3100
		UHF Communication Systems
A22.		C-51-475-000/MS-001 Antenna Coupler Group OE-5012
A23.		C-51-475-E00/MS-001 7000 Channel control module
A24.		C-51-547-000/MS-001 RT-5085 Maint Manual Vol 1

A25.		C-51-547-000/MS-002 RT-5085 Maint Manual Vol 2
A26.		C-51-559-000/MY-001 RT-5073(V)1/WSC-3 IPB and Schematic, Vol 3
A27.		C-51-559-000/MF-001 RT-5073 (V)1/WSC-3 Operation and Maintenance Vol 1
A28.		C-51-559-000/MF-001 RT-5073 (V)1/WSC-3 Schematics, Vol 3
A29.		C-51-559-000/MN-001 RT-5073 (V)1/WSC-3 Maintenance with Parts List, Vol 2
A30.		C-51-559-000/NY-001 RT-5073(V)/WSC-3 Preventive Maintenance
A31.		C-51-560-000/NK-001 RT-5073 (V)/WSC-1 Preventive Maintenance
A32.		C-51-560-000/NY-001 RT-5073 (V)/WSC-1 Preventive Maintenance
A33.		C-51-561-000/MS-001 Receiver Radio R-5113/SRR
A34.		C-51-561-000/NK-001 Receiver Radio R-5113/SRR Performance Test Sheet
A35.		C-51-564-000/MS-001 Antenna AS-5169/SRC

A36.		C-51-565-000/MS-001 AS-5170/SRC, Antenna, P/N 451233
A37.		C-51-574-000/MS-001 UHF LOS Radio Communication System
A38.		C-51-702-000/MS-001 Operation and Maintenance Manual with Parts List for Control, Radio Set C-5576/SSC
		HF Transmit
A39.		C-51-573-000/MS-001 AN/SRC-513 Radio set B band (HF)
A40.		C-51-584-F00/MS-001 F/5094/UR Band past Filter
A41.		C-51-588-000/MS-001 HF Transmitter Subsystem, AN/SRT-504, Maintenance Pub.
A42.		C-51-631-000/MS-001 Performance Monitor RF-1815A Instruction Manual.
A43.		C-51-632-000/MS-001 Power Amplifier RF-1110A and Power Supply RF-1124
A44.		C-51-633-000/MS-001 RF1310-13 HF Exciter
A45.		C-51-639-000/MS-001 Operator Alarm Panel, BZ-5021/U

A46.		C-51-665-000/MS-001 Power Amplifier RF-1110B (AM-5334/UR), Instruction Manual
A47.		C-51-665-000/MS-001 AM/5334/UR RF Amp
A48.		
A49.		C-51-666-000/MS-001 F-5094/UR Bandpass Filter
A50.		C-51-667-000/MS-001 ID-5138/SR Transmit System Monitor
A51.		C-51-668-000/MP-001 Bandpass Filter F-5088/SR, Repair and Overhaul Specification
A52.		C-51-669-000/MS-001 CU-5165/UR Output Combiner
A53.		C-51-670-000/MS-001 CD-5021/UR HF Transmit System Control Unit
A54.		C-51-673-000/MS-001 OE-5034/SR 35’ Whip Antenna
A55.		C-51-675-000/MS-001 OE-5035/SR 18’ Whip Antenna
A56.		C-51-677-000/MS-001 Lightweight Broadband HF Communications Antenna Series (LWCA) AS-5199/U
		LF/MF/HF Receive

A57.		C-51-568-000/MS-001 HF Receiver RF-590, Instruction Manual
A58.		C-51-568-000/MS-Z01 RF-592, RX Remote Control Assembly
A59.		C-51-568-000/MS-Z02 RF-553-01, Remote Preselector Interface
A60.		C-51-568-000/MS-Z03 RF-555-02, ISB Option
A61.		C-51-597-000/MB-001 VLF/LF Receiver Type GS7110, Operating Manual
A62.		C-51-597-000/MS-001 VLF/LF Receiver Type GS7110, Maintenance Manual with Illustrated Parts List
A63.		C-51-677-000/MS-001 Lightweight Broadband HF Communications Antenna Series (LWCA), Installation and Maintenance
A64.		C-51-677-000/MS-Z01 Lightweight Broadband HF Communications Antenna Series (LWCA) AS-5199/U
A65.		C-51-678-000/MS-001 Filter Bandpass F-5082/A/URC and F-5089/UR
A66.		C-51-681-000/MS001 AS-5199/SR Fan Antenna
A67.		C-51-681-000/MS-001 AS-5196/U HF RX Antenna

A68.		C-51-682-000/MS-001 CU-5130/U LF/MF/HF Multicoupler
A69.		C-51-684-000/MS-001 Switch, Radio Frequency Transmission Line SA- 5153/WSC, Technical Manual
A70.		C-51-685-000/MS-001 SB-5154/SRC HF Crosspoint Matrix/Patch Panel
A71.		C-51-699-000/MS-001 MD-5077(V)3/U PSK Modem
A72.		C-51-682-000/MS-001 Broadband Multicoupler (CU-5130)
A73.		C-51-406-000/MS-001 Marine Digital Watchkeeping Receiver Type WK2182C
		CCMS
A74.		C-51-563-000/MS-001 CCMS Supervisor's Console
A75.		C-51-570-000/MS-001 Regulated Power Supplies MLGS-5A Series
A76.		C-51-653-000/MS-001 Operation and Maintenance Manual with Parts List CCMS Supervisor Console
A77.		C-51-575-000/MS-001 Communication Control and Monitoring System (CCMS) 2 Books, Vol 1 and 2

A78.		C-51-578-000/MS-001 Control Monitor Set, AN/SYM-501 (CCMS)
A79.		C-51-586-000/MS-001 J5224/SRC UHF LOS Patch Panel
A80.		C-51-639-000/MS-001 UHF Distress Monitor Panel, LS-5026/SRC
A81.		C-51-638-000/MS-001 J-5223 UHF LOS Antenna Patch Panel
A82.		C-51-639-000/MS-001 Operator Alarm Panel, BZ-5021/U
A83.		C-51-640-000/MS-001 CW Keyer Panel TH-5011/UGC
A84.		C-51-649-000/MS-001 Processor Select Panel, SB-5155/U, Preventive Maintenance
A85.		C51-652-000/MS-001 CCMS Operator Console
A86.		C-51-702-000/MS001 C-5576/SSC EMCON Panel, Preventive Maintenance (RTCU)
A87.		C-51-703-000/MS-001 C-5577/SSC Radio Transmit Control Unit (RTCU)
A88.		C-51-654-000/MS-001 Control and Monitoring System Operators Manual

		OE-82C
A89.		C-51-528-000/MP-001 Antenna Group OE-82C/WSC-1 (V) Overhaul and Repair
A90.		C-51-586-000/MS-001 AN/SSC-510 UHF Satellite Radio Communication System
A91.		C-51-528-000/MS-001 Antenna Group OE-82C/WSC-1 (V) Operation and Maintenance
A92.		C-51-528-000/NY-001
A93.		C-51-528-000/NY-002
A94.		C-51-528-000/NY-Z01
A95.		C-51-528-000/NK-001
A96.		C-51-528-000/NK-002
		DAMA
A97.		C-51-883-000/MS-001
A98.		C-51-883-000/MS-002

A99.		C-51-883-000/MB-001
A100.		C-51-883-000/MB-002
A101.		C-51-883-000/NY-001
A102.		C-51-883-000/NY-Z01
		FBB
A103.		C-51-958-000/MS-001
A104.		C-51-958-000/MS-002
A105.		C-51-958-000/MS-003
A106.		C-51-958-000/MS-004
A107.		C-51-958-000/MB-001
A108.		C-51-958-000/MB-002
A109.		C-51-958-000/MB-003
A110.		C-51-958-000/MB-004

A111.		C-51-958-000/NY-001
		CMS
A112.		C-60-236-000/MD-001
A113.		C-60-236-000/MM-001
A114.		C-60-236-000/NK-001
A115.		C-60-236-000/NY-001
		SHINCOM 3100
A116.		C-52-476-000/MS-001 TECHNICAL MANUAL CPF SHINCOM 3100
A117.		C-52-476-000/MB-004 Operator's Guide Recorder Storage Unit Single Interface Node
A118.		C-52-476-000/MB-007 SHINCOM 3100-Technical Manual Automatic Software Upgrade (ASU) Console Application
A119.		C-52-476-000/MB-003 Operator's Guide Helicopter Audio Distribution System Single Interface Node

A120.		C-52-289-001/MY-001 Shipboard Integrated Interior Communications System AN/SSC-505(V), Parts Identification List
A121.		C-52-476-000/MB-001 OEPRATOR'S GUIDE SHINCOM 3100 DUAL SCREEN TERMINAL
A122.		C-52-476-000/MB-002 Outdoor Terminal Operator’s Guide
A123.		C-52-476-000/MB-005 SHINCOM 3100 – PUBLIC ADDRESS SINGLE INTERFACE NODE OPERATOR’S GUIDE
A124.		C-52-476-000/MB-006 Operator’s Guide Console Dual Screen Terminal
A125.		C-52-476-000/MS-002 TECHNICAL MANUAL CONSOLE DUAL SCREEN TERMINAL
A126.		C-52-476-000/NK-001 NAVAL PREVENTIVE MAINTENANCE PERFORMANCE TEST COMMUNICATIONS) INTERIOR COMMUNICATIONS SYSTEM (SHINCOM) 3100
A127.		C-52-476-000/NT-001 READINESS TRIAL FOR INTERIOR COMMUNICATIONS SYSTEM (SHINCOM) 3100
A128.		C-52-476-000/NY-001 NAVAL PREVENTIVE MAINTENANCE SCHEDULE (COMMUNICATIONS) INTERIOR COMMUNICATIONS SYSTEM (SHINCOM) 3100

A129.		C-52-289-001/MS-001 DESCRIPTION AND MAINTENANCE INSTRUCTIONS - SHIPBOARD INTEGRATED INTERIOR COMMUNICATIONS SYSTEM AN/SSC- 505(V) BOOK 1, 2, 3

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task	PO
Task Area I	**Maintain Internal Communications System**	
I001	Initialize an Internal Communication System to an operational state. (e.g. SHINCOM, ICS, VICES)	
I002	Initialize an Internal Communication System to an operational state. (e.g. SHINCOM, ICS, VICES)	
I003	Install a new or repaired/refurbished/overhauled Internal Communication System component	
I004	Monitor an Internal Communication System performance	

I005	Operate an Internal Communication System	
I006	Perform an Internal Communication System functional check	
I007	Perform an Internal Communication System PM activity (including a test/trial/inspection)	
I008	Analyze/evaluate/interpret an Internal Communication System functional check result	
I009	Analyze/evaluate/interpret an Internal Communication System test/trial/inspection result or report	
I010	Perform post-repair calibration/testing/tuning/alignment of an Internal Communication System	
I011	Remove a faulty component from an Internal Communication System	
I012	Bench test a repaired/refurbished/overhauled Internal Communication System component	
I013	Calibrate/test/tune/align a newly installed Internal Communication System component	
I014	Conduct preliminary fault diagnostic on an Internal Communication System (Tech of the Watch)	
I015	Configure an Internal Communication System to meet an operational requirement	
Duty Area J	**Maintain UHF Communication Systems**	
J001	Initialize an Ultra High Frequency (UHF) System to an operational state	

J002	Install a new or repaired/refurbished/overhauled UHF System component	
J003	Monitor a UHF System performance	
J004	Operate a UHF System	
J005	Perform a UHF System functional check	
J006	Perform a UHF System PM activity (including a test/trial/inspection)	
J007	Perform post-repair calibration/testing/tuning/alignment of a UHF System	
J008	Remove a faulty component from a UHF System	
J009	Calibrate/test/tune/align a newly installed UHF System component	
J010	Conduct preliminary fault diagnostic on a UHF System (Tech of the Watch)	
J011	Configure a UHF System to meet an operational requirement	
J012	Diagnose a UHF System fault to the Lowest Repairable Unit	
Duty Area J	**Maintain HF Transmit System**	
J013	Initialize a High Frequency (HF)/ Medium Frequency (MF)/ Low Frequency (LF)/ Very Low Frequency (VLF) Transmit System to an operational state	

J014	Install a new or repaired/refurbished/overhauled HF/MF/LF/VLF Transmit System component	
J015	Monitor an HF/MF/LF/VLF Transmit System performance	
J016	Operate an HF/MF/LF/VLF Transmit System	
J017	Perform an HF/MF/LF/VLF Transmit System functional check	
J018	Perform an HF/MF/LF/VLF Transmit System PM activity (including a test/trial/inspection)	
J019	Perform post-repair calibration/testing/tuning/alignment of an HF/MF/LF/VLF Transmit System	
J020	Remove a faulty component from an HF/MF/LF/VLF Transmit System	
J021	Calibrate/test/tune/align a newly installed HF/MF/LF/VLF Transmit System component	
J022	Conduct preliminary fault diagnostic on an HF/MF/LF/VLF Transmit System (Tech of the Watch)	
J023	Configure an HF/MF/LF/VLF Transmit System to meet an operational requirement	
J024	Diagnose an HF/MF/LF/VLF Transmit System fault to the Lowest Repairable Unit	
Duty Area J	**Maintain LF/MF/HF Receive System**	
J025	Initialize a High Frequency (HF)/ Medium Frequency (MF)/ Low Frequency (LF)/ Very Low Frequency (VLF) Receive System to an operational state	

J026	Install a new or repaired/refurbished/overhauled HF/MF/LF/VLF Receive System component	
J027	Monitor an HF/MF/LF/VLF Receive System performance	
J028	Operate an HF/MF/LF/VLF Receive System	
J029	Perform an HF/MF/LF/VLF Receive System functional check	
J030	Perform an HF/MF/LF/VLF Receive System PM activity (including a test/trial/inspection)	
J031	Perform post-repair calibration/testing/tuning/alignment of an HF/MF/LF/VLF Receive System	
J032	Remove a faulty component from an HF/MF/LF/VLF Receive System	
J033	Calibrate/test/tune/align a newly installed HF/MF/LF/VLF Receive System component	
J034	Conduct preliminary fault diagnostic on an HF/MF/LF/VLF Receive System (Tech of the Watch)	
J035	Configure an HF/MF/LF/VLF Receive System to meet an operational requirement	
J036	Diagnose an HF/MF/LF/VLF Receive System fault to the Lowest Repairable Unit	
	Maintain the UHF Satellite Communication System	
J037	Initialize an Ultra High Frequency (UHF) System to an operational state	

J038	Install a new or repaired/refurbished/overhauled UHF System component	
J039	Monitor a UHF System performance	
J040	Operate a UHF System	
J041	Perform a UHF System functional check	
J042	Perform a UHF System PM activity (including a test/trial/inspection)	
J043	Perform post-repair calibration/testing/tuning/alignment of a UHF System	
J044	Remove a faulty component from a UHF System	
J45	Calibrate/test/tune/align a newly installed UHF System component	
J46	Conduct preliminary fault diagnostic on a UHF System (Tech of the Watch)	
J47	Configure a UHF System to meet an operational requirement	
J48	Diagnose a UHF System fault to the Lowest Repairable Unit	
Duty Area K	**Maintain the Naval Information System**	
K013	Initialize a HSDC/MCOIN/COWAN System to an operational state	
K014	Install a new or repaired/refurbished/overhauled HSDC/MCOIN/COWAN System component	

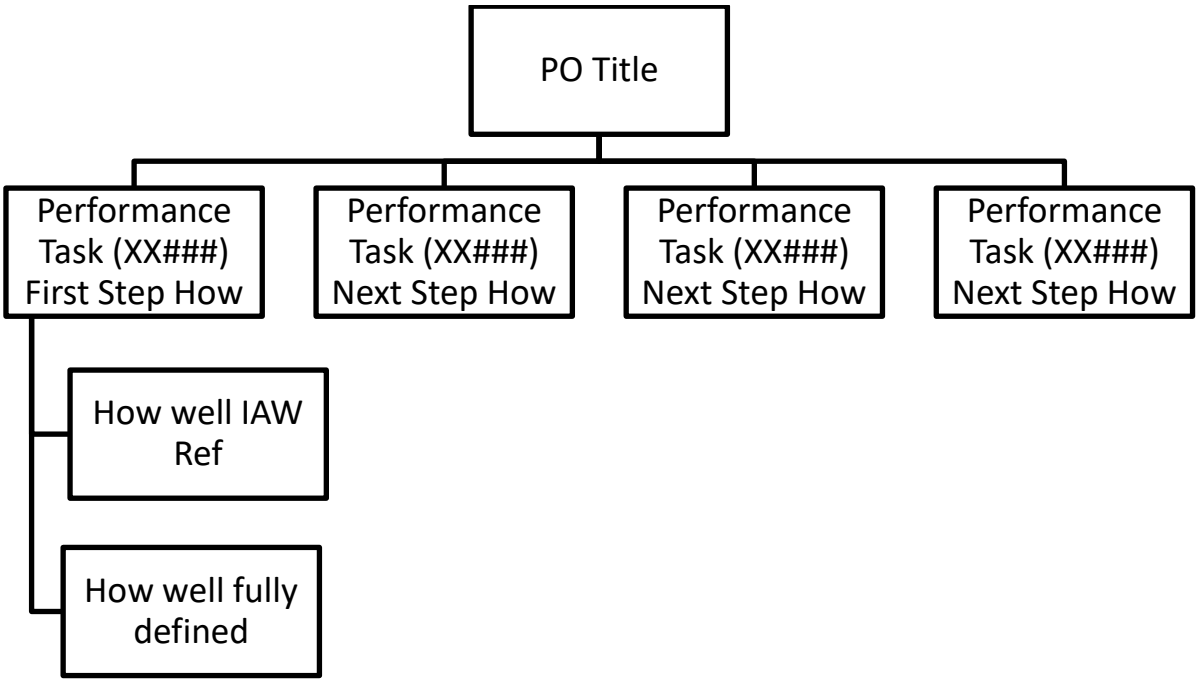
K015	Monitor a HSDC/MCOIN/COWAN System performance	
K016	Operate a HSDC/MCOIN/COWAN System	
K017	Perform a HSDC/MCOIN/COWAN System functional check	
K018	Perform a HSDC/MCOIN/COWAN System PM activity (including a test/trial/inspection)	
K019	Perform post-repair calibration/testing/tuning/alignment of a HSDC/MCOIN/COWAN System	
K020	Remove a faulty component from a HSDC/MCOIN/COWAN System	
K021	Calibrate/test/tune/align a newly installed HSDC/MCOIN/COWAN System component	
K022	Conduct preliminary fault diagnostic on a HSDC/MCOIN/COWAN System (Tech of the Watch)	
K023	Configure a HSDC/MCOIN/COWAN System to meet an operational requirement	
K024	Diagnose a HSDC/MCOIN/COWAN System fault to the Lowest Repairable Unit	
Duty Area L	**Maintain Communications Control and Monitoring System**	
L01	Initialize a Communication Control and Monitoring System (CCMS) to an operational state	
L02	Install a new or repaired/refurbished/overhauled CCMS component	

L03	Operate a CCMS	
L04	Perform a CCMS functional check	
L05	Perform a CCMS PM activity (including a test/trial/inspection)	
L06	Perform post-repair calibration/testing/tuning/alignment of a CCMS	
L07	Remove a faulty component from a CCMS	
L08	Calibrate/test/tune/align a newly installed CCMS component	
L09	Conduct preliminary fault diagnostic on a CCMS (Tech of the Watch)	
L010	Configure a CCMS to meet an operational requirement	
L011	Diagnose a CCMS fault to the Lowest Repairable Unit	
Duty Area O	**Maintain the Tactical Data System**	
O001	Initialize a Link System to an operational state	
O002	Install a new or repaired/refurbished/overhauled Link System component	
O003	Monitor a Link System performance	
O004	Operate a Link System	
O005	Perform a Link System functional check	

O006	Perform a Link System PM activity (including a test/trial/inspection)	
O007	Perform post-repair calibration/testing/tuning/alignment of a Link System	
O008	Remove a faulty component from a Link System	
O009	Calibrate/test/tune/align a newly installed Link System component	
O010	Configure a Link System to meet an operational requirement	
O011	Diagnose a Link System fault to the Lowest Repairable Unit	
Duty Area O	**Combat Management system 330**	
O012	Initialize a Command and Control System (CCS) to an operational state	
O013	Install a new or repaired/refurbished/overhauled CCS component	
O014	Monitor a CCS performance	
O015	Operate a CCS	
O016	Perform a CCS functional check	
O017	Perform post-repair calibration/testing/tuning/alignment of a CCS	
O018	Remove a faulty component from a CCS	

O019	Calibrate/test/tune/align a newly installed CCS component	
O020	Conduct preliminary fault diagnostic on a CCS (Tech of the Watch)	
O021	Configure a CCS to meet an operational requirement	
O022	Diagnose a CCS fault to the Lowest Repairable Unit	
	DRMIS	
	As Per DRMIS QSP	

ANNEX C – QS SCALAR



ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 25 May 10 to 4 June 10
Location: S37 Classroom B17 Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
STONIER	P.R.	CPO1	00119	D MAR PERS 3-2- 3
MAGEE	M.G.	CPO1	00119	DMTE
SEVIGNY	L.A.P.	CPO2	00119	FMF Cape Scott
SOBOTKIEWICZ	D.	PO1	00117	NFS CSE DIV
MAYBURRY	J.D.	PO1	00117	CFFS(E) CSE DIV
NICHOLSON	M.L.	PO2	00117	NFS CSE DIV
ROEBUCK	G.B.	PO2	00117	HMCS HALIFAX
MACNEIL	E.M.	MS	00117	NFS CSE DIV
PASK	D.J.	Lt(N)	00211	CFNOS STDS DIV
MURRAY	D.R.	CIV	EL-07	NFS ERT DIV
MITCHELL	D.	CIV	EL-07	NFS ERT DIV
HOGENBIRK	D.S.	ASLt	00344	NFS OT DIV

Board Objective:

1. To create the Job Qualification for the W Eng Communication System

Maintainer. Specification Task – No Train Decisions:

2. The tasks selected for training are detailed at Annex B. Tasks not selected for training are listed below with substantiation:

TASK SERIAL	TASK ELEMENT	SUBSTANTIATION

Discussions / Decisions:

DAY 1 – 25 MAY 10

1. Introductions

2. Briefed on philosophy of new training scheme

3. Reviewed/modified equipment list:

4. Because RDU is part of OJT, WADS will be made a teaching point in CCMS

5. MHS designated OJE

6. CRYPTO designated OJE

7. CCTV designated OJE

8. SHINCOM switch was added as formal training, as it is not covered in any previous training

9. Reviewed chapter 1

10. Created PO breakdown of chapter 2

11. Reviewed chapter 3

12. Reviewed Annex K

13. It was decided in a previous board that CCS would be instructed to all system maintainers because of its minimal training time and the benefit it would provide all system maintainers with regards to ERT

DAY 2 – 26 MAY 10

1. SHINCOM

2. Created Standards

3. Created Teaching Points

4. Created EO 1.01 – Describe SHINCOM

5. Created EO 1.02 – Maintain SHINCOM

6. The board decided that flight-deck broadcast will be included in the PACOM OJT at the LS RQ

DAY 3 – 27 MAY 10

1. Revised PO 2 – UHF

2. Created EO 2.01 – Describe UHF Communication System

3. Created EO 2.02 – Initialize UHF

4. Created EO 2.03 – Maintain UHF Transceiver

5. Created EO 2.04 – Maintain UHF Coupler Group

6. The board decided that UHF and VHF standalone systems will be learned under OJE.

7. Created PO 3 – HF Transmit

8. Created EO 3.01 – Describe HF Transmit System

9. Created EO 3.02 – Initialize HF Transmit System

10. Created PO 4 – LF/MF/HF Receive

11. The board decided to introduce LF/MF here because it was not included in previous training

12. The board decided that the HF backpack will be OJE training.

DAY 4 – 28 MAY 10

1. Completed EO 3.03 – Maintain ISDU and TSCU

2. Completed EO 3.04 – Maintain HF Transmit System

3. Completed EO 4.01 – Describe LF/MF/HF Receive System

4. Completed EO 4.02 – Initialize LF/MF/HF Receive System

5. Completed EO 4.03 – Maintain LF/MF/HF Receive System

6. Completed EO 5.01 – Describe CCMS

7. Completed EO 5.02 – Initialize CCMS

8. Started EO 5.03 – Manipulate CCMS software

9. Completed EO 5.04 – Maintain the CCMS

10. It was decided that CCMS would be taught in 2 EOs: one for software manipulation and one for hardware.

DAY 5 – 31 MAY 10

1. Created PO 6 Maintain UHF Satellite Communication System

2. Completed EO 6.01 - Describe the UHF Satellite Communication System

3. Completed EO 6.02 - Initialize the UHF Satellite Communication Systems

4. Completed EO 6.03 - Maintain the OE-82C

5. Completed EO 6.04 - Maintain the Demand Assign Multiple Access System

6. Completed EO 6.05 - Maintain the ON-143

7. Created PO 7 - Maintain Satellite Communication System

8. Completed EO 007.01 - Describe the NERA SATURN-Bm

9. Completed EO 007.02 - Initialize the NERA SATURN-Bm

10. Completed EO 007.03 - Maintain the NERA SATURN-Bm

11. Separate POs were created for UHF Satellite Communication System and Satellite Communication System because NERA-Bm training was too large for an EO.

12. Completed EO 5.03 – Manipulate CCMS software

13. Revised PO 008 - Maintain Naval Information System

14. Completed EO 008.01 - Describe the High Speed Data Connectivity System

15. Completed EO 008.02 - Initialize the High Speed Data Connectivity System

16. Completed EO 008.03 - Maintain the High Speed Data Connectivity System

17. SECLAN, subnet relay, and MINeMS were considered for this QSP, but the board decided that this training will be added when equipment is available and approved for training.

18. Revised PO 009 - Maintain Tactical Data Systems

19. Completed EO 009.01 - Describe the Link System

20. Completed EO 009.02 - Initialize the Link System

21. Completed EO 009.03 - Maintain the Link System

22. Completed EO 009.04 - Describe the Global Command and Control System Maritime

23. Completed EO 009.05 - Initialize the Global Command and Control System Maritime

24. Completed EO 009.06 - Maintain the Global Command and Control System Maritime

25. The board decided to only implement PCs because each PO is manageable enough that ECs are not necessary.

DAY 6 – 1 JUNE 10

1. During the UHF EO's, CCMS is not involved with the fault finding. The board discussed adding UHF and HF system faults during the CCMS EO's. The board decided that UHF and HF system faults will be introduced during the CCMS EO's, and the student will be expected to identify faults.

2. Assigned the number of periods for:

3. EO 001.01 - Describe the SHINCOM switch

4. EO 001.02 - Maintain the SHINCOM Switch

5. EO 002.01 - Describe the UHF Communication Systems

6. EO 002.02 - Initialize the UHF Communication System

7. EO 002.03 - Maintain the UHF Transceivers

8. EO 002.04 - Maintain the UHF Coupler Group

9. EO 003.01 - Describe the HF Transmit System

10. EO 003.02 - Initialize the HF Transmit System

11. EO 003.03 - Maintain the ISDU and TSCU

12. EO 003.04 - Maintain the HF Transmit System

13. EO 004.01 - Describe the LF/MF/HF Receive System

14. EO 004.02 - Initialize the LF/MF/HF Receive System

15. EO 004.03 - Maintain the LF/MF/HF Receive System

16. EO 005.01 - Describe the Communications Control and Monitoring Systems

17. EO 005.02 - Initialize the Communications Control and Monitoring Systems

18. EO 006.01 - Describe the UHF Satellite Communication System

19. EO 006.02 - Initialize the UHF Satellite Communication Systems

20. EO 006.03 - Maintain the OE-82C

21. EO 006.04 - Maintain the Demand Assign Multiple Access System

22. EO 006.05 - Maintain the ON-143

23. EO 007.01 - Describe the NERA SATURN-Bm

24. EO 007.02 - Initialize the NERA SATURN-Bm

25. EO 007.03 - Maintain the NERA SATURN-Bm

DAY 7 – 2 JUNE 10

1. TDO recommended that EO 009.02 be made a teaching point under EO 009.03 as it is only one period. The board understood the reasoning but preferred to have the QSP closely mirror other W Eng QSPs at this level.

2. TDO recommended that EO 009.05 be made a teaching point under EO 009.06 as it is only one period. The board understood the reasoning but preferred to have the QSP closely mirror other W Eng QSPs at this level.

3. Reviewed and assigned the number of periods for

4. PO 001 - Maintain Internal Communication Systems

    a. EO 001.01 - Manipulate the SHINCOM switch

    b. EO 001.02 - Maintain the SHINCOM switch

5. PO 002 - Maintain UHF Communication Systems

    a. EO 002.01 - Describe the UHF Communication Systems

b. EO 002.02 - Initialize the UHF Communication System

c. EO 002.03 - Maintain the UHF Transceivers

d. EO 002.04 - Maintain the UHF Coupler Group

6. PO 003 - Maintain HF Transmit System

a. EO 003.01 - Describe the HF Transmit System

b. EO 003.02 - Initialize the HF Transmit System

c. EO 003.03 - Maintain the ISDU and TSCU

d. EO 003.04 - Maintain the HF Transmit System

7. PO 004 - Maintain the LF/MF/HF Receive System

(1) EO 004.01 - Describe the LF/MF/HF Receive System

(2) EO 004.02 - Initialize the LF/MF/HF Receive System

(3) EO 004.03 - Maintain the LF/MF/HF Receive System

8. PO 005 - Maintain Communications Control and Monitoring Systems

a. EO 005.01 - Describe the Communications Control and Monitoring Systems

b. EO 005.02 - Initialize the Communications Control and Monitoring Systems

c. EO 005.03 - Manipulate Communications Control and Monitoring System software

d. EO 005.04 - Maintain the Communications Control and Monitoring System

9. PO 006 - Maintain UHF Satellite Communication System

a. EO 006.01 - Describe the UHF Satellite Communication System

b. EO 006.02 - Initialize the UHF Satellite Communication Systems

c. EO 006.03 - Maintain the OE-82C

d. EO 006.04 - Maintain the Demand Assign Multiple Access System

e. EO 006.05 - Maintain the ON-143

10. PO 007 - Maintain Satellite Communication System

a. EO 007.01 - Describe the NERA SATURN-Bm

b. EO 007.02 - Initialize the NERA SATURN-Bm

c. EO 007.03 - Maintain the NERA SATURN-Bm

11. PO 008 - Maintain Naval Information System

a. EO 008.01 - Describe the High Speed Data Connectivity System

b. EO 008.02 - Initialize the High Speed Data Connectivity System

12. PO 009 - Maintain Tactical Data Systems

a. EO 009.01 - Describe the Link System

b. EO 009.02 - Initialize the Link System

c. EO 009.03 - Maintain the Link System

d. EO 009.04 - Describe the Global Command and Control System Maritime

e. EO 009.05 - Initialize the Global Command and Control System Maritime

f. EO 009.06 - Maintain the Global Command and Control System Maritime

13. PO 010 - Maintain Command and Control System

a. EO 010.01 - Describe the Command and Control System Function

b. EO 010.02 - Initialize the Command and Control System

14. The board discussed the HSDC mock-up at the schools and feels that this is not sufficient to properly represent the ships fitted system. The board feels that students do not get proper training on the HSDC system. The board recommends that a scaled down version of the HSDC system be implemented at the school.

15. The board discussed the issue of rebuilding GCCS software. The board included this rebuild in the training because of operator inexperience.

DAY 8 – 3 JUNE 10

1. Completed Annex A
2. Updated reference lists in Chapter 2 and Chapter 4
3. Completed Annex B, Annex F, Annex G, Annex H

DAY 9 – 4 JUNE 10

1. The TDO recommended that all tasks involving description be removed from the task list. DPGR recommended at this time to leave description in the task list as

it has not been officially decided how knowledge will be classified in the job-based specification.

2. The TDO recommended that references should be aligned with teaching points. The board decided that this was not necessary as all CFTOs are structured identically and the publication layout is covered in the first EO of each PO.

3. Board conducted a general review of the

document Lessons Learned:

1. For future boards, it is recommended that the first PO to be designed fall into the generic technicians model and not be the exception (i.e. SHINCOM)

2. Although a generic template was used for the teaching points, there needs to be a correlation between the standard statement and the teaching points.

3. Guidance to Instructional Designers:

4. Ideally, all practice fault-finding is conducted in pairs. To conduct one practice fault the board allotted 4 hours for a course of 8. If 20 periods are allotted for practice faults, the board intention is to have 5 practice faults for that period. If time allows, the instructors are free to insert additional faults, or consider individual faults.

5. The level of detail taught on any specific PO should be commensurate with the time allotted.

Follow-up Action:

1. DMTE to create and insert task lists. On completion, DMTE to distribute completed QSP to board members, TDO, and supervisors for approval.

Other Considerations (if applicable)

1. N/A

TRAINING PLAN ANNEXES

ANNEX E – ON JOB PERFORMANCE RECORD

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

3. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

5. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

7. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

8. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

9. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

10. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

11. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

12. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Projector, Smart board etc.)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**

	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	XXX
ATT Sub-total	**XXX**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	XXX
Training Administrative Time (TAT)	XXX
Training Travel Time (TTT)	XXX
Physical Fitness (PT)	XXX
Sub-total	**XXX**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**XXX**

ANNEX G.1 – OPTIMUM TRAINING SCHEDULE

Week 1 Time	Day 1	Day 2	Day 3	Day4	Day 5
Period 1 **0800-0850**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 2 **0900-0950**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 3 **1000-1050**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 4 **1100-1150**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 5 **1300-1350**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 6 **1400-1450**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 7 **1500-1550**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 8	**EO/EC /PC/Admin** Title(s) / Activity				

Week 1 Time	Day 1	Day 2	Day 3	Day4	Day 5
1600-1650					

ANNEX H – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	When	Resources	Pass Mark	Impact
PC 001	EO 001.01 and EO 001.02	To evaluate the Trainees ability to Maintain Internal Communications System	Practical	O/C of EO 001.02	Internal Communications System	60%	Re-attempt TRB
PC 002	EO 002.01 Thru EO 002.04	To evaluate the Trainees ability to Maintain the UHF Communications System	Practical	O/C of EO 002.04	UHF Communications System	60%	Re-attempt TRB
PC 003	EO 003.01 Thru EO 003.04	To evaluate the Trainees ability to Maintain the HF Transmit System	Practical	O/C of EO 003.04	HF transmit System	60%	Re-attempt TRB
PC 004	EO 004.01 Thru EO 004.03	To evaluate the Trainees ability to Maintain the LF/MF/HF Receive System	Practical	O/C of EO 004.03	LF/MF/HF Receive System	60%	Re-attempt TRB
PC 005	EO 005.01 Thru EO 005.04	To evaluate the Trainees ability to Maintain the CCMS	Practical	O/C of EO 005.04	CCMS	60%	Re-attempt TRB

PC 006	EO 005.01 Thru EO 006.05	To evaluate the Trainees ability to Maintain the UHF Satellite Communications	Practical	O/C of EO 006.05	Satellite Communications System	60%	Re-attempt TRB
EC 007.01	EO 007.01	To evaluate the Trainees ability to Initialize the FBB	Practical	O/C of EO 007.01	FBB	60%	Re-attempt TRB
PC 007	EO 007.01 Thru EO 007.03	To evaluate the Trainees ability to maintain the Satellite Communications System	Practical	O/C of EO 007.03	Satellite Communications System	60%	Re-attempt TRB
PC 009	EO 009.01 Thru EO 009.03	To evaluate the Trainees ability to maintain the Tactical Data System	Practical	O/C of EO 009.05	Tactical Data System	60%	Re-attempt TRB
EC 009.04	EO 009.04	To evaluate the Trainees ability to describe the DLPS	Written	O/C of EO 009.04	DLPS references	60%	Re-attempt TRB
EC 009.05	EO 009.05	To evaluate the Trainees ability to initialize the DLPS	Written	O/C of EO 009.05	DLPS Publications	60%	Re-attempt TRB

EC 009.06	EO 009.06	To evaluate the Trainees ability to repair the DLPS	written	O/C of EO 009.06	DLPS Publications	60%	Re-attempt TRB
EC 009.07	EO 009.07	To evaluate the Trainees ability to conduct PM on the DLPS	Written	O/C of EO 009.07	DLPS Publications	60%	Re-attempt TRB
PC 009	EO 009.04 Thru EO 009.07	To evaluate the Trainees ability to maintain the Tactical Data System	Practical	O/C of EO 009.07	DLPS	60%	Re-attempt TRB
PC 010	EO 010.01Thru EO 010.02	To evaluate the Trainees ability to maintain the CMS	Practical	O/C of EO 010.03	CMS	60%	Re-attempt TRB
PC 011	IAW PO 002 – Utilize DRMIS Portal as a Maintenance Supervisor (Mod 2) QSP119097						Re-attempt TRB

ANNEX I – ASSESSMENT CHECKLISTS

PART I - ADMINISTRATIVE INSTRUCTIONS

1. This is a generic list for administering a PC/EC.
2. The Assessor shall:
    a. ensure all publications, test equipment and equipment are serviceable or Trainee informed of deficiencies;
    b. ensure the Trainees observe all safety precautions;
    c. test Trainees individually;
    d. supervise and provide physical assistance. as required;
    e. provide a pre and post assessment waiting area; and
    f. review Annex K with the Trainee.
3. If, after allotted time, the Trainee is showing sufficient progress the Assessor may allot 15 additional minutes to complete the PC/EC with the loss of two points.
4. For each additional 5 minutes after the initial 15 minutes will result in two points being lost. The maximum extra time granted will be 30 minutes. If the student does not find the fault after the additional 30 minutes, the instructor will stop the PC/EC and calculate the final mark.
5. The Assessor is to ensure that the best use of time allotted for PC's/EC's is made by facilitating the PC/EC process. For example:
    a. if the Trainee is about to perform a logical but time consuming procedure the Assessor should supply the results and have the Trainee resume without penalty to the student however, if the action is not justified by the Trainee it is considered incorrect procedure; or
    b. after a maximum of 10 minutes of inactivity the Assessor shall provide assistance.
6. Pass mark is 60%

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. You are given the following:
    a. References;

b. Course notes;

c. Job sheets;

d. Work sheets;

e. Projects; and

f. Equipment and/or MPT.

2. IAW the references and PC/EC list, you shall:

a. Power up;

b. Conduct system functional checks;

c. Load software or crypto as required;

d. Localize fault to LRU;

e. Repair;

f. Verify operational readiness;

g. Report status to assessor; and

h. Power down.

3. Prior to performing a test or measurement you shall inform the Assessor of your intention.

4. IAW the references and PC/EC List, you shall complete the PC/EC:

a. Within allotted time IAW Chapter 4;

b. Individually;

c. With supervision; and

d. Physical assistance. as required.

5. IAW the references, you shall observe all safety precautions. A safety infraction or violation will result in instructor intervention and the instructor may terminate PC/EC based on the severity of the violation.

a. 12. Pass mark is 60%.

b. 13. Do you have any questions?

Trainee Sign: _____Date: _____

Instructor Sign: _____Date: _____

PC/EC EVALUATION SHEET			
Equipment:		**Name:**	
PO Number:	**Start:**	**Stop:**	**Total:**
Course:			
Note: When allocating a score use only whole numbers. Circle the appropriate number in score column			

Initialization (10%)		
Score	**Justification**	**Remarks**
1 - 2	Did not follow correct flash up procedure. Flash up faults not recognized	
3 - 4	Followed procedure but flash up faults not recognized	
5 - 6	Followed flash up procedure. After some hesitation, corrected flash up faults	
7 - 8	Followed Flash Up procedure. Recognized and corrected flash up faults	
9 - 10	Confidently Flashed Up, efficiently corrected flash up faults.	

Use of Test Equipment (10%)		
Score	**Justification**	**Remarks**

1 - 2	Did Not select appropriate TE for Fault/ Did not demonstrate correct TE operating procedures	
3 - 4	Hesitant with selection of TE/Unsure of TE set up	
5 - 6	Selected correct TE/ Demonstrate weak operating procedures	
7 - 8	Select Correct TE/Properly utilized TE	
9 - 10	Selected Correct TE considering best options/Demonstrated thorough knowledge of operation	
Fault Finding Procedures (30%)		
Score	**Justification**	**Remarks**
1 - 2	Hesitant to begin Fault Finding./Did not recognize fault indications	
3 - 4	Did not correctly interpret Fault Indications; BITE; Test Equipment Readings; Took the wrong path and did not recover.	

5 - 6	Minor inconsistencies in interpreting fault indications. May have taken wrong path but recovered. Employed logical faultfinding	
7 - 8	Recognized and Correctly interpreted fault indications. Employed Logical Faultfinding.	
9 - 10	Efficiently interpreted Fault Indications and evaluated all symptoms. Found fault with minimum number of steps.	
System Knowledge (30%)		
Score	**Justification**	**Remarks**
1 - 2	System knowledge did not support faultfinding. Could not answer questions from instructor. Could not properly use supporting documents	
3 - 4	System knowledge weak. Could not answer all questions. Struggled with documentation to support faultfinding. Could not fully explain fault	
5 - 6	System knowledge weak in some areas. Answered most questions from instructor and gave general explanations regarding the fault.	

7 - 8	Good system knowledge. Use of supporting documentation aided in fault finding. Explained fault clearly.	
9 - 10	Detailed system knowledge. Used documentation efficiently to find fault. Gave detailed explanation of fault	
Timing (10%)		
Score	**Justification**	**Remarks**
10	Found Fault in allotted time	
8	Found Fault with an extra 15 minutes	
6	Found Fault with an extra 20 Minutes	
4	Found Fault with an extra 25 Minutes	
2	Found Fault with an extra 30 Minutes	
Find the Fault (10%)		
Score	**Justification**	**Remarks**
10	Fault Corrected	

Score	**Weighting**	**Total**
Initialization	**X 1**	**%**
Test Equipment	**X 1**	**%**

Fault Finding	**X 3**	**%**
System Knowledge	**X 3**	**%**
Timing	**X 1**	**%**
Finding the Fault	**X 1**	**%**
	TOTAL	**%**

Assessor Signature:	**Trainee Signature:**

ANNEX J – ON JOB PERFORMANCE RECORD

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 1.0	DD / MM/ YYYY	New OJPR

Name and Initials	Rank	Service Number

Date Training Started	DD MMM YYYY
Date of Completion	DD MMM YYYY

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers).

Example OJPR record:

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI and Unit
PC/EC XXX.XX				DD/MM/ YYYY	

OJPR Reviewed By OJT Supervisor (Rank, Position, Name)	Signature	Date
Section Head (NCMs) or HOD (Officers)		

TRAINING COMPLETION CERTIFICATE

Date training completed:	
Div Officer (NCMs) or CO (Officers)	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Name of QSP.

TE Cmdt/CO	Date

ANNEX K – TT PLAN

PO/EO	OBJECTIVE	EO	EC	PC	PO
PO 001	Maintain Internal Communication Systems	30		8	38
EO 001.01	Manipulate the SHINCOM 3100	14			
EO 001.02	Maintain the SHINCOM 3100	16		8	
PO 002	Maintain UHF Communication Systems	56		8	64
EO 002.01	Describe the UHF Communication Systems	5			
EO 002.02	Initialize the UHF Communication System	4			
EO 002.03	Maintain the UHF Transceivers	21			
EO 002.04	Maintain the UHF Coupler Group	26		8	
PO 003	Maintain HF Transmit System	58.5		8	66.5
EO 003.01	Describe the HF Transmit System	10			
EO 003.02	Initialize the HF Transmit System	3.5			
EO 003.03	Maintain the ISDU and TSCU	22			
EO 003.04	Maintain the HF Transmit System	23		8	
PO 004	Maintain the LF/MF/HF Receive System	15.5		8	23.5
EO 004.01	Describe the LF/MF/HF Receive System	6.5			

EO 004.02	Initialize the LF/MF/HF Receive System	1			
EO 004.03	Maintain the LF/MF/HF Receive System	8		8	
PO 005	Maintain Communications Control and Monitoring Systems	70		8	78
EO 005.01	Describe the Communications Control and Monitoring Systems	16			
EO 005.02	Initialize the Communications Control and Monitoring Systems	6			
EO 005.03	Manipulate Communications Control and Monitoring System software	28			
EO 005.04	Maintain the Communications Control and Monitoring System	20		8	
PO 006	Maintain UHF Satellite Communication System	46		8	54
EO 006.01	Describe the UHF Satellite Communication System	7			
EO 006.02	Initialize the UHF Satellite Communication Systems	4			
EO 006.03	Maintain the OE-82C	15			
EO 006.04	Maintain the Demand Assign Multiple Access System	20		8	
PO 007	Maintain Satellite Communication System	155		24	179
EO 007.01	Describe the FBB	14			

EO 007.02	Initialize the FBB	1			
EO 007.03	Maintain the FBB	8		8	
EO 007.04	Describe the MSCU	21			
EO 007.05	Initialize the MSCU	5.5			
EO 007.06	Maintain the MSCU	39.5		8	
EO 007.07	Describe the NMT	21			
EO 007.08	Initialize the NMT	5.5			
EO 007.09	Maintain the NMT	39.5		8	
PO 008	Not Allocated				
PO 009	Maintain Tactical Data Systems	58	4	16	78
EO 009.01	Describe the Link 11 System	9			
EO 009.02	Initialize the Link 11System	3			
EO 009.03	Maintain the Link System	8		8	
EO 009.04	Describe the DLPS	10	1		
EO 009.05	Initialize the DLPS	10	1		
EO 009.06	Repair the DLPS	16	1	8	
EO 009.07	Conduct PM for the DLPS	2	1		

PO 010	Maintain CMS 330	2030		8	38
EO 010.01	Describe the CMS330	10			
EO 010.02	Initialize the CMS 330	10			
EO 010.03	Maintain the CMS 330	10		8	
PO 011	Utilize DRMIS Portal as a Maintenance Supervisor	5			5

* Includes debrief time.

ANNEX L – RECORD OF DECISIONS TPWP

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title				
Location of Board				
Dates				
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair (content)				
Co-Chair (process)				
NFS				
SME				
SME				
SME				
SME				
SME/Secretary				
Advisor				

Day #	Details	Action
	The following information on target population was compiled during the initial steps of the TPWB to give context for design and development: (rank, experience, entry level education, etc.).	
1		
2		

ANNEX L.1 – RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Amendment Form, thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	
Trg Mgr:	

ANNEX M – ACRONYMS / ABBREVIATIONS

Acronym / Abbreviation	Term / Definition
MPT	Maintenance Procedural Trainer - Can be any form of computer based training, table top faults, or any other form of fault finding practice necessary.
SHIPLAN	Ship Local Area Network
HSDC	High Speed Data Connectivity
LAN	Local Area Network
NOC	Network Operations Centre
EMCON	Emission Control
DWAN	Defence Wide Area Network
IOS	Internal Operating System
MCOIN	Maritime Command Operational Information Network
CFE	Centrix Four Eyes
GCCS-M	Global Command and Control System Maritime
DNS	Digital Network Server
UPS	Uninterruptable Power Supply
VOIP	Voice Over Internet Protocol

DPE	Digital Processing Equipment
COMFAC	Communications Facility
CCS	Command and Control Systems
CCMS	Communications, Control and Monitoring System
BITE	Built In Test Equipment
RIMM	Removable Internal Mass Media
SCI	Status Control Indicator
RTCU	Radio Transmit Control Unit
ECS	External Communication System